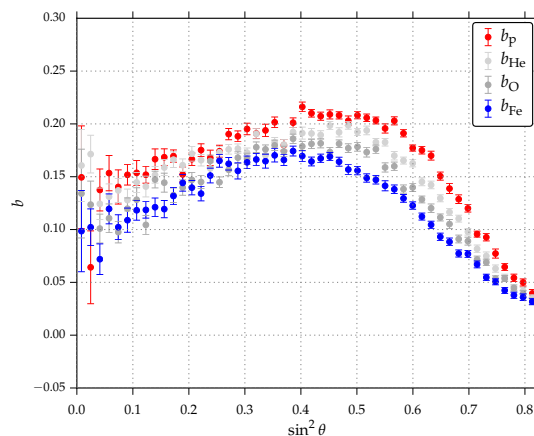


Asymmetries of the Lateral Distribution of Particles at the Ground

(Asymmetrien der lateralen Verteilung von Teilchen
am Erdboden)

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I declare that I have developed and written the enclosed thesis completely by myself, and have not used sources or means without declaration in the text.

Karlsruhe, 01.06.2018

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Introduction

Cosmic Rays (CR), i.e. highly energetic charged and uncharged particles, photons and neutrinos coming from the space, enter the Earth's atmosphere continuously from all directions. Inside the atmosphere they produce cascades of subparticles, the so called air showers, through various interacting processes. Due to energies exceeding a certain limit, these subparticles are able to reach the ground.

The CR were first discovered in the early 20th century during balloon flights. Their discovery caused a new episode in high-energy physics and in the following decades helped to develop new possibilities of measurements in astrophysics. The energy range of CR turned out to be between 10^9 eV and several 10^{20} eV. The formation and acceleration processes of lower energetic particles (up to 10^{16} eV) are today well understood. However, particles were detected with much larger energies up to 10^{20} eV, Ultra-High Energy Cosmic Rays (UHECR). Their origin so far remains an unsolved problem. To explain this enigma, various theories have been developed, but have to be validated experimentally.

For this purpose, and since cosmic radiation contains a lot of information about various astronomic objects and phenomena, the Pierre Auger Observatory, an international research institution located in Argentina, has been built. It is, as yet, the largest CR observatory in the world. Its surface detector, a grid of water-Cherenkov detectors spread over a huge area in the Argentinian Pampa, is able to measure the "footprint" of an air shower. For one single event, i.e. one cascade of particles hitting the detector grid, only a few of detectors near the center of the air shower, the shower core, obtain signal, and most of the others do not. Therefore, the result is a very narrow and distinct area around the core in which signal is measured.

Here is where the thesis' main objects lie. The measured air-shower "footprints" are not perfectly symmetric shaped, but have some asymmetric features. Many different effects, like projection and other geometric effects or an attenuation of charged particles, cause this asymmetry. Within this thesis we first develop a simplified theoretical model to parametrize this signal asymmetry and to linearize it for easier understanding. In the second part, an empirical parametrization is made out of simulated data considering the results of the linearized model. This parametrization could then be used in future improvements of the reconstruction of air showers by the Pierre Auger Observatory to counterbalance the so far occurring core shift. Additionally, it will potentially improve the angular resolution of the arrival direction of the CRs.

This thesis is structured in the following way. In Chapter 1 some general features of CR and their propagation are explained. Additionally, general and technical information about the Pierre Auger Observatory is given and important abbreviations are introduced. Chapter 2 aims for developing a simplified air shower model and linearizing the resulting signal formula. Finally, in Chapter 3 simulation data of the simulation library NAPOLI were analyzed to develop an empirical function of the asymmetry of the lateral distribution of CR on the ground with respect to the different asymmetry-causing effects. In the end the results are discussed and a prospect for further studies is given.

Chapter 1

Theoretical Background

1.1 Cosmic rays

In this first Chapter we explain general characteristics of CR, their sources and their acceleration mechanisms. Additionally, we introduce some important terms and abbreviations in context of cosmic radiation and the Pierre Auger Observatory.

At first, one distinguishes between primary cosmic rays and secondary cosmic rays in two different manners. On the one hand, in context of the propagation of CR, the particles coming directly from their sources are called primary particles. The secondary particles are those which are produced by interaction of primaries with interstellar matter. On the other hand, the terms primary and secondary particles are used in context of their detection on the earth. Here, primaries are the high-energy particles coming from space and colliding with the atmosphere. Secondaries are the particles arising through interaction of primaries (and other secondaries) with the atmosphere and also the particles detected on the ground. The interaction between primaries and matter is very complex. There are many processes of interaction and therefore a lot of secondary particles arise. In the atmosphere these interaction processes generate cascades of secondary particles which are called Extensive Air Showers (EAS). In Section 1.1.2 this phenomenon is explained in more detail. The Pierre Auger Observatory explores and analyzes such particle cascades and therefore contributes to the determination of origins of the cosmic rays.

Furthermore, one differentiates between galactic and extragalactic cosmic radiation, i.e. the particles produced inside the Milky Way and in the other galaxies. This distinction is important to understand the characteristics of the all particle energy spectrum, which is discussed in Section 1.1.1.

1.1.1 Primary cosmic rays

Primary cosmic radiation contains charged particles like protons or electrons as well as uncharged particles like neutrinos and photons. The Pierre Auger Observatory is intended to analyze the charged component of the cosmic radiation, because it is easier to detect than uncharged particles. Thus in the following the focus will be on the charged CR.

Characteristics

The charged component of primary cosmic radiation is composed of small ionized nuclei and leptons (electrons and muons). In energy regions higher than several GeV the main constituents are protons ($\approx 85\%$) and α particles ($\approx 12\%$), whereas ionized atoms with a nuclear charge of ≥ 3 are subdominant ($\approx 3\%$) as well as electrons ($\approx 0.8\%$). [1]

In Figure 1.1.1 the all-particle energy spectrum (starting at 10^{13} eV) of the CR is presented. It has been shown that this spectrum has a power-law behaviour proportional to $E^{-\gamma}$ with γ

CHAPTER 1. THEORETICAL BACKGROUND

the spectral index. There are some characteristics of the spectrum which are known as “knee”, “ankle” and “toe”. These are explained in the following [2–4]. Since the energy spectrum of CR is object of recent research, not all characteristics are well understood. Especially, in the very high energetic region, $E > 10^{19}$ eV, only assumptions have been made for the spectrum’s behaviour. Among other research institutions, the Pierre Auger Observatory collects data to find answers to these unsolved problems.

Knees. The first knee, i.e. the first sharp change of the spectral slope, occurs in an energy range of 2 to 5 PeV. At this point, the spectral index changes from $\gamma \approx 2.7$ to $\gamma \approx 3.0$, which effects a steepening of the curve. The second knee occurs at approx. 400 PeV, where γ becomes approx. 3.3. There are two different assumptions which might be the reasons for the spectrum behaviour. The first explanation deals with the measured fact that the intensity of light CR ($Z < 6$) shows a break in this energy range. For protons this break lies just in the energy range of the knee, for iron it is in the range of the second knee. Since the main constituent of cosmic radiation are protons, the first knee is more dominant than the second. The reason for the intensity break of light CR is that the CR sources reach their maximum acceleration energy. Due to the fact that heavier particles can be accelerated to higher energies, the bump in the spectrum moves from the energy range of the first knee to the range of the second for increasing Z .

The second explanation is based on the fact that at energy levels counting more than 1 PeV the particles are able to leave the galaxy. This is associated with the Larmor radius which is again correlated with Z . So, dependent on Z , more and more particles leave the galaxy and the measured particle flux decreases more steeply.

Although nowadays the first explanation is seen as the more probable one, it is not totally sure whether the second explanation or a mixture of both is more accurate.

Ankle. Another special point in the all-particle spectrum is the so called “ankle”. It appears at an energy of about 5 EeV. Here the spectral index γ decreases back to about 2.7. The ankle is explained by a crossover of the source position from galactic to extragalactic regions. That means, above an energy of 5 EeV particles from other galaxies have enough energy to reach the Milky Way and the Earth. So, the flux of particles increases which implies a decrease of γ .

Note here that the experimental verification of the theory that the ankle is a consequence of the transition from galactic to extragalactic sources is one of the first big achievements of the Pierre Auger Observatory.

Suppression. Above $10^{19.5}$ eV the all-particle spectrum shows a sharp suppression. The flux of particles decreases to zero almost instantly. Since the data of the Pierre Auger Observatory are statistically reliable, the cut off is seen as verified [5]. However, as yet its reason is still unclear. A former theory, the GZK-theory named after the physicists Greisen, Zatsepin and Kuzmin, states that the proton dominated CR interact with the Cosmic Microwave Background (CMB) as long as the proton energy is smaller than 5×10^{19} eV. However, the Auger data shows [6–10] that in high energy region the CR are not dominated by protons anymore but by heavier nuclei. That led to a falsification of the GZK-theory. To date, various theories have been postulated. The current research intends to falsify or verify these theories. One concept which matches the data from the Pierre Auger Observatory postulates a maximum energy particles can achieve in extragalactic accelerators.

Acceleration mechanisms

After the consideration of the energy spectrum of CR the question emerges about how the charged particles can be accelerated to energies above 10^{20} eV. The simplest way to accelerate charged particles is with electric fields. In fact, in some regions near black holes or neutron stars there are electric fields of that magnitude where particles can be accelerated to the highest energies. However, in large parts of the space, plasmas destroy the strong electric fields. On the contrary to

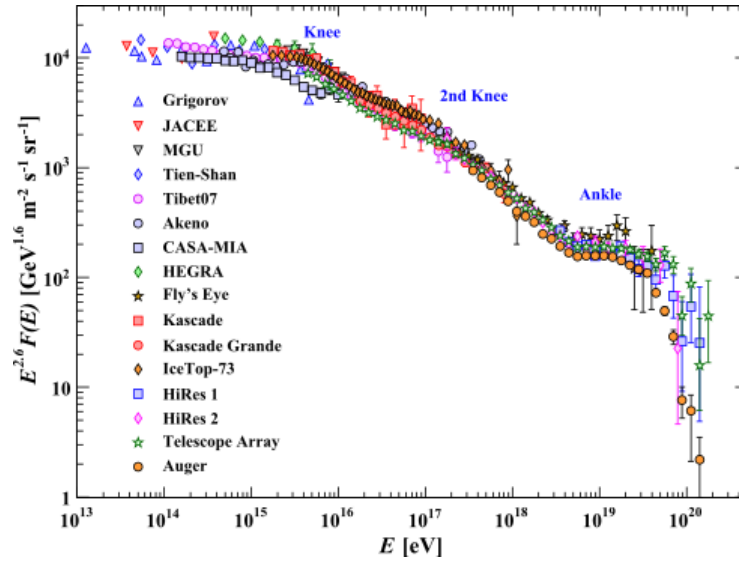


Figure 1.1.1: All-particle spectrum. When plotting the particle flux (scaled with $E^{2.6}$) over the particle energy this power-law curve with various features typically arises. The logarithmic scaling highlights the knee and the ankle. [11]

the electric fields, huge magnetic fields are omnipresent throughout the universe. Those varying magnetic fields, which imply again electric fields, can be seen as the more common accelerators of cosmic rays. In the following, the two most popular acceleration mechanisms based on magnetic fields are briefly discussed, the *Fermi-acceleration* and *Unipolar Inductors*. [12]

The *Fermi-acceleration* is based on the transfer of energy from the motion of a macroscopic object, like an interstellar cloud (*second order Fermi-mechanism*) or a shock wave (*first order*), to microscopic particles. In the second order Fermi-mechanism, the particles receive their energy increase from the scattering on interstellar matter. In the first order Fermi-mechanism, the shock wave acts like a magnetic mirror, where the particles are reflected back and forth several times and every reflection leads to an energy gain. Since the first order mechanism is more efficient, the high-energy CR are assumed to have their large energies from this acceleration mechanism. Additionally, the shock waves of supernovae are believed to be the dominant accelerators of high-energy galactic CR whereas the shock waves of gamma-ray bursts (GRB), Active Galactic Nuclei's jets and hot spots (AGN) and gravitational accretion are the popular region of UHECRs origin.

In contrast to the Fermi acceleration, the model of *Unipolar Inductors* is based on the huge electric fields induced by extremely fast magnetic rotators like neutron stars or black holes. The combination of giant rotational energy and magnetic fields causes relativistic outflows like "winds" and jets whose large electric fields are able to accelerate particles to very high energies.

The possible sources can be summarized in the so called *Hillas Plot*, Figure 1.1.2, which is a useful visualization of the capability of sources. There, the magnetic field of a source candidate is plotted with respect to its size including uncertainties. One can see that, as yet, neutron stars, AGN jets, and GRBs are the most probable sources of UHECRs.

1.1.2 Secondary cosmic rays and extensive air showers

The previous Chapter is about particles directly coming from their sources after being accelerated. However, the particles being detected on Earth are not these primaries, but particles which are products of several interaction and perturbation processes of the primary with atmospheric nuclei. This statement can be justified by the fact, that the radiation length for photons and electrons in air is $X_0 = 36.66 \text{ g/cm}^2$ and the interaction length for hadrons in air is $\lambda = 90.0 \text{ g/cm}^2$ [1], whereas the atmospheric depth is about 1000 g/cm^2 . So, the depth of the atmosphere is equivalent to

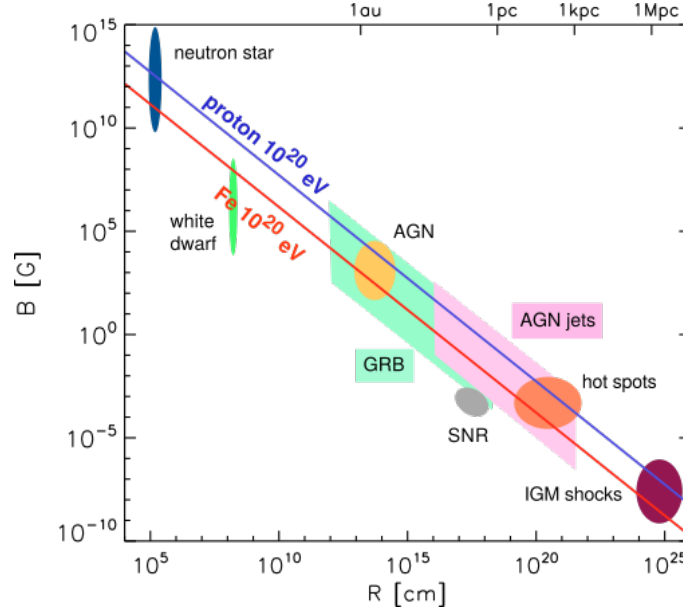


Figure 1.1.2: In the Hillas plot the magnetic flux density B is plotted against the size R of the various astrophysical candidates for UHECR sources. The two diagonal lines represent the borders of CR energy. The sources lying above these borders are able to accelerate protons (blue) and iron (red) up to an energy of 10^{20} eV. [12]

about 27 radiation lengths and 11 hadronic interaction lengths. This implies that not a single original primary reaches the sea level. Furthermore, there is another consequence of this fact: the incoming primary particle has to possess a minimum energy (of about 10^{10} eV) so that at least a few secondaries arrive at the sea level. In order to get sufficient statistics the number of detected particles has to be large enough, so that the primary energy has to be higher than 10^{13} eV. This is the reason why the all-particle spectrum starts at energies of about 10^{13} eV and not lower.

As it was written earlier in this Chapter, one single primary particle causes a cascade of secondaries through various interaction processes. Such a cascade, the so called Extensive Air Shower, travels mostly in direction of the incoming primary particle. However, after several interaction distances, it receives more and more of the transverse momentum, so that the lateral distribution grows. Although, the size of the shower front does not rise continuously, but, through different attenuation effects of charged particles, it shrinks after a specific distance covered. The altitude on which the shower reaches its maximum is an important value for analyzing CR. It corresponds to a particular grammage the particles have traveled so far. The grammage at the shower maximum altitude is called $X_{\max}$ and is usually measured in g/cm^2 .

The shower itself consists of mainly three different components, the electromagnetic, the muonic, and the hadronic component. These are named after their constituents (see Figure 1.1.3). There are two additional components in EASs, the neutrinos and the radiation part, where the neutrinos are linked to the muonic and the photons to the electromagnetic shower. However, the SD at the Pierre Auger Observatory (see Chapter 1.2.2) cannot observe these two uncharged components. In the following subsections the three main shower components are presented. [4]

Hadronic component

The hadronic component plays an important role in the early shower development. Although the total number of hadronic particles arising in an EAS averages out at about only 1 %, these hadrons are the origin of the other shower components (see also Figure 1.1.3).

The initial primary particle collides with an atmospheric nucleus under the strong interaction,

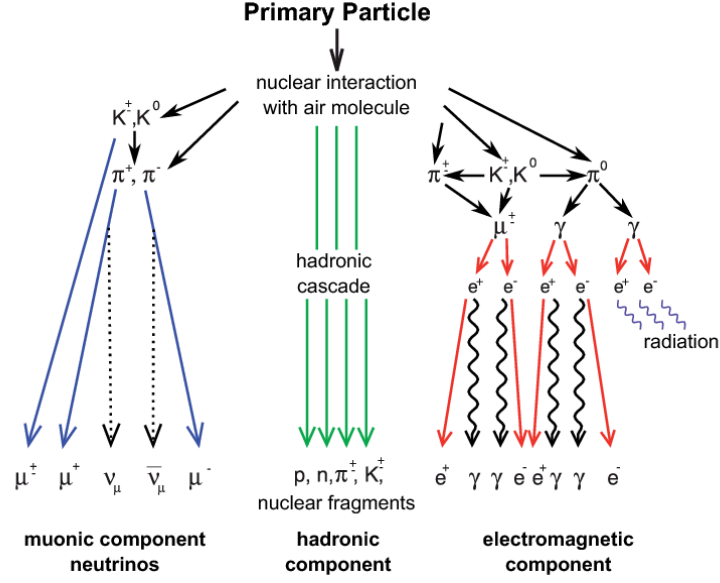


Figure 1.1.3: Components of an EAS. The main constituents are the electromagnetic particles (electrons, positrons, photons), the hadrons (protons, kaons, pions), and the muons. [13]

the first generation of secondaries. Subsequently, these secondary hadrons either collide with other air nuclei or decay into leptons, neutrinos, and photons. As a consequence, a hadronic cascade arises mainly consisting of charged and neutral pions, followed by kaons, protons, and neutrons and the other shower components are initiated.

A characteristic of the hadronic shower component is the small lateral distribution. Since the traverse momenta of the hadrons can be nearly neglected compared to the longitudinal momenta, the hadronic component hits the ground within a radius of about 20 m around the shower axis.

Muonic component

The main source of muons in an EAS is the decay of charged kaons and pions, the products of the hadronic shower component. The three processes,

$$\begin{aligned}\pi^\pm &\longrightarrow \mu^\pm + \nu_\mu (\bar{\nu}_\mu), \\ K^\pm &\longrightarrow \mu^\pm + \nu_\mu (\bar{\nu}_\mu), \\ K^\pm &\longrightarrow \pi^0 + \mu^\pm + \nu_\mu (\bar{\nu}_\mu),\end{aligned}$$

are the dominating mechanisms producing muons. One can see that also neutrinos are generated here. Regarding the total number of generated particles in an EAS, the muons account for approx. 5 %.

The muons mainly arise at an early stage of the shower development, since the deeper in the atmosphere, the higher is the air-particle density and as a consequence the higher is the possibility of interaction between hadrons and air nuclei while muons are not decaying. Besides, the muons are not effected by strong interaction, so only ionization can lower their energy. This causes a nearly unperturbed flight to the ground. In addition to the fact that a muon is mainly emitted in the same direction as its generating hadron, this enables a precise reconstruction of the early secondary hadrons. Additionally, the number of muons increases with atmospheric depth to a certain maximum (dependent on the primary particle and its energy) and because of the very low interaction rate this maximum number nearly stays unchanged until the shower reaches the ground. Therefore, the muonic signal measured with the Surface Detector of the Pierre Auger Observatory is large.

The electromagnetic component

The electromagnetic component is the dominating one in an EAS. It accounts for approx. 95 % of the total number of generated particles. Within the electromagnetic cascade, photons are the prevailing component with approx. 80 %, followed by electrons and positrons. The bulk of photons is created through the decay of neutral pions,

$$\pi^0 \longrightarrow 2 \gamma .$$

Only a minority of photons arises through a decay of muons. Now, the photons produce pairs of electrons and positrons,

$$\gamma + \text{nucleus} \longrightarrow \text{nucleus} + e^+ + e^- ,$$

which again generate photons via bremsstrahlung,

$$e^\pm + \text{nucleus} \longrightarrow \text{nucleus} + e^\pm + \gamma .$$

Hence, many electromagnetic subcascades are induced by repeating these processes several times. In contrast to muons, electrons (and positrons) interact with the atmosphere which implies a loss of particles in dependence on atmospheric depth. It turns out that the total number of electrons shows a distinctive maximum. Now, this maximum leads to the definition of X_{max} , which was introduced in the beginning of the Subsection 1.1.2. X_{max} is now the grammage at that altitude where the electromagnetic shower component reaches its maximum of total particle number.

1.2 Pierre Auger Observatory

The Pierre Auger Observatory [14], located in the Pampa Amarilla (Argentina), is a large research institution detecting cosmic radiation and was built in order to explore especially more about UHECR. The Observatory is run by an international collaboration of scientists. It consists on the one hand of 1660 water-Cherenkov detectors, evenly spread over an area of about 3000 km². This is the so called Surface Detector (SD) (details in Section 1.2.2). On the other hand the Observatory has 27 Fluorescence Detectors (FD) (details in Section 1.2.1) distributed in four different places around the SD area. The SD enables to determine the energies of secondary particles very precisely, whereas the FD helps to detect the trajectories. Because of the combination of these two detection methods, the Observatory is a hybrid detector, which allows very detailed and precise observation of UHECR. In Figure 1.2.1 the detector area of the Pierre Auger Observatory is shown.

1.2.1 Fluorescence detector

Since this thesis focuses on the SD measurement, only a brief overview of the FD is given. Through interaction of charged particles in an air shower with atmospheric nitrogen, UV light is emitted by the so called fluorescence effect. The intensity of this light changes proportionally to the number and the energy of the charged particles [15]. Therefore, the development of an air shower can be reconstructed by detecting the fluorescence light. At the Pierre Auger Observatory the FDs were built for this reason. One single fluorescence telescope consists of a grid of focusing mirrors (3.6 m × 3.6 m, spherical) which reflects the light onto a grid of cameras (80 cm × 80 cm). These cameras are very sensitive, which allows a usage of the FDs in very dark, moonless nights only. In Figure 1.2.2 there are two sketches of the FD structure.

1.2.2 Surface detector

As it was written in the introduction part of this Chapter, the SD is a huge grid of single water-Cherenkov detectors spread over an area of about 30 times the size of Paris. These detectors are basically 12,000-liter water tanks filled with highly purified water, shielded from any light

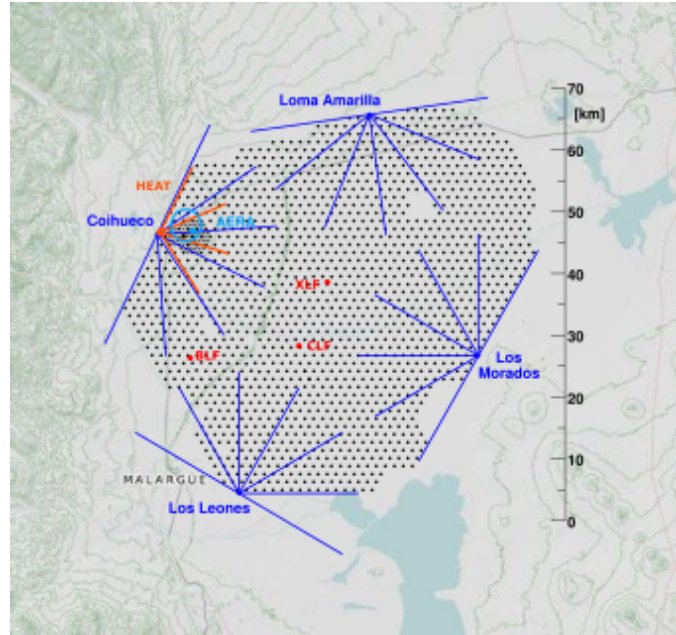


Figure 1.2.1: Map of the Pierre Auger Observatory. Each black dot corresponds to one of the SD tanks. Additionally, the four sites of the FD houses are shown with the field of view of each six telescopes. The Coihueco site hosts three additional high elevation telescopes (HEAT). [6]

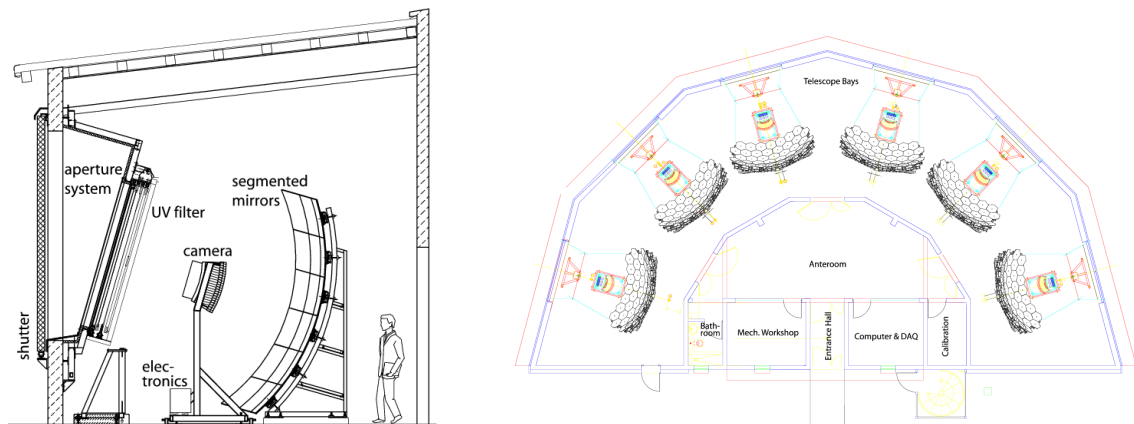


Figure 1.2.2: Setup of the Fluorescence Detector. On the *left* side one can see the structure of one telescope: Through an aperture and filter system the fluorescence light shines on the mirror grid from which it is reflected to the camera. The picture on the *right* shows one of the four FD houses which contain six telescopes each. [16]

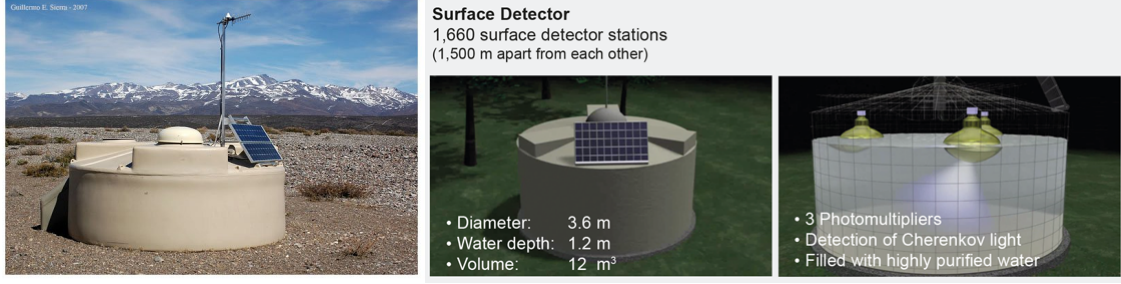


Figure 1.2.3: (Left:) Photo of a single water-Cherenkov station, (right:) Set up of the Surface Detector. The picture shows a computer graphic of one station and an insight into it. The dimensions of the station are shown as well as the three PMTs mounted on the detector. [14]

from outside. Most of the detector stations are laid out on a triangular grid of 1.5 km spacing. In a smaller sub-part of the SD area, the Infill area, the density of tanks is higher (750 m to each neighbour). Here a measurement for lower energies is possible.

The measurement process is the following: when a charged particle enters the SD station with an energy so that it travels faster than the speed of light in water, the so called Cherenkov light is emitted by the electromagnetic shockwave of the particle. This light is captured by three very sensitive photomultiplier tubes (PMT) which convert it into electric signals. Since the track length of the incoming secondary particle in water is dependent on its energy, it is possible to reconstruct its energy by analysing the Cherenkov light signal.

Each EAS consists of billions of secondary particles which hit the ground spread over an area with a diameter of a few kilometers. So, for one single shower multiple stations yield almost simultaneous signal. Typically, a cosmic ray with energy of 10^{19} eV arriving vertically causes signal in 8 stations [17]. Out of this observed amount of data collected by multiple stations, the primary energy can be reconstructed. Furthermore, through the small time differences between the light detection of several stations, the determination of the primary particle arrival direction is achieved with an angular accuracy of about 1° [17]. To guarantee that the shower core is within the active SD array, there is a pre-selection of the detected events. The station gaining the highest signal must be encased by six stations. Within this active hexagon, all six stations have to be operational at the time of the event [5]. In Figure 1.2.3 one SD station is shown. In contrast to the FD, the SD has the advantage of continuous operation. However, out of financial reasons the SD stations are spread in intervals of 1.5 km, not smaller. Thus, the SD can directly measure only a part of the shower.

To sum it up, the SD measures energies of secondary shower particles on the ground. This means that in a way the SD measures the footprint of an EAS. Now, for the reconstruction of the primary, the detected signal S has to be parametrized as a function of several parameters, e.g. the distance of the active station to the shower core or the inclination of the shower. Here, some effects cause asymmetries of signal measurement that would naively not be expected. Asymmetry and processes that cause it will be described in the next Chapter.

Chapter 2

Origin of Signal Asymmetry

For rotationally symmetric and vertical showers we naively expect concentric circles as equal signal lines around the shower core for SD measurement. And also for a non vertical shower, we expect highly-symmetric equal-signal lines, i.e. ellipses, with the core as their center. However, the real “footprints” of EASs are not that symmetric. Core shifts and other asymmetries arise which have various reasons. The main causes generating asymmetry are the shape of the detectors, attenuation effects of charged particles, and the underlying geometric model of an EAS. These phenomena are explained in the following.

Note here that the term “footprint” of an air shower means the equal-signal contour lines produced by an EAS at the ground. It is a cross section of the shower with the ground plane. It thus depends on the geometrical shape of the shower.

2.1 Shower coordinate system

Before starting further discussions of asymmetry effects, a reference coordinate system is defined (see Figure 2.1.1). The z -axis points along the shower axis $\hat{a}$. The plane generated by the x - and y -axis is perpendicular to it. The point where the shower axis hits the ground, the shower core point, is set as origin of the reference coordinate system. In this coordinate system, the shower axis representation gets the components $\hat{a} = (0, 0, 1)$. The normal to the ground is $\hat{n} = (-\sin \theta, 0, \cos \theta)$, where θ is the zenith angle of the shower. The y -axis is chosen to be perpendicular to both, $\hat{n}$ and $\hat{a}$. Hence, the y -axis lies in the ground plane and the x -axis points into the ground. All in all, the right-handed reference coordinate system is thus defined by the three orthogonal unit vectors

$$\hat{e}_z = \hat{a}, \quad \hat{e}_y = [\hat{n} \times \hat{a}], \quad \hat{e}_x = \hat{e}_y \times \hat{e}_z, \quad (2.1)$$

where $[\hat{x}] = \vec{x}/|\vec{x}|$.

A vector starting at the origin of the reference coordinate system and ending in any point in the ground plane is denoted with $\vec{\rho}$. It can be set to a position of any SD station. Its parallel projection into the xy -shower-plane is given by $\vec{r} = r(\cos \psi, \sin \psi, 0)$ with the azimuth angle ψ in the shower plane measured relatively to the x -axis $\hat{e}_x$. By definition, the upstream stations are located in the direction with $\psi = 0^\circ$, the downstream stations with $\psi = \pm 180^\circ$, and the “neutral” stations lie in directions with $\psi = \pm 90^\circ$. The vector $\vec{\rho}$ is a projection of $\vec{r}$ along the shower axis $\hat{a}$ and thus $\vec{\rho} = \vec{r} + k \hat{a}$, but also lies in the ground plane and thus $\vec{\rho} \cdot \hat{n} \equiv 0$. The unknown k can be calculated from $\hat{n} \cdot (\vec{r} + k \hat{a}) = 0 \Rightarrow k = -(\hat{n} \cdot \vec{r})/(\hat{n} \cdot \hat{a})$ and therefore

$$\vec{\rho} = \vec{r} - \frac{\hat{n} \cdot \vec{r}}{\hat{n} \cdot \hat{a}} \hat{a}. \quad (2.2)$$

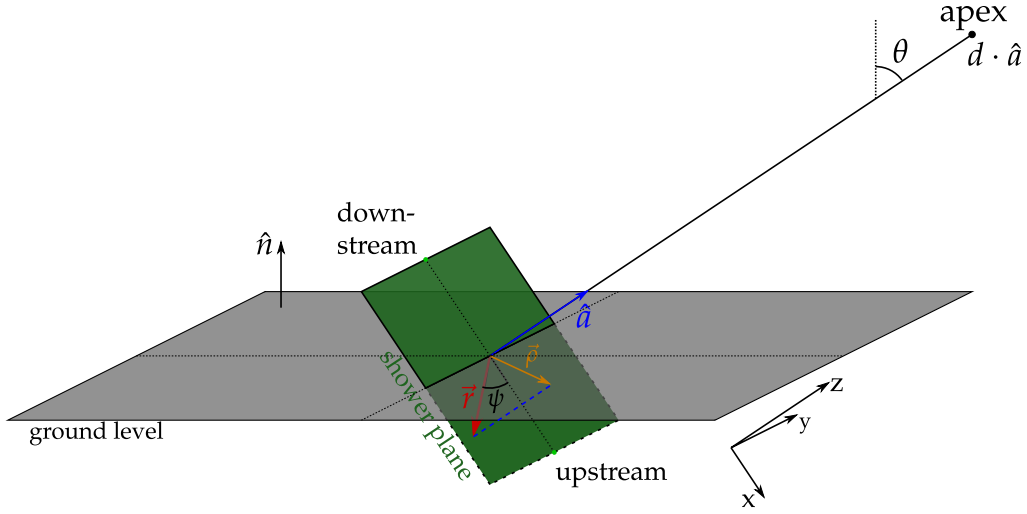


Figure 2.1.1: This figure shows all relevant parameters for the parametrization of the projection of an inclined air shower in coordinates of the shower plane.

2.2 Geometrical model of an extensive air shower

As it was shortly introduced above, referring to the issue of asymmetry in signal measurement of the SD, the geometrical model of an air-shower shape becomes very important. This is shown in the following with the help of two simple geometric models. On the one hand, an axisymmetric shower is considered, which is a suitable model for very distant showers. However, this is a very unrealistic model for closer showers. Therefore, it is only discussed briefly (Section 2.2.1). On the other hand, the model of a conical shower, with its apex at the shower origin, is assumed. This is a very simple model, too, but closer to the reality than the axisymmetric one. The real longitudinal shower shape is described by the Gaisser-Hillas function [18]. Since the conical is a more realistic model, it is the basis for all calculations of the signal asymmetry in Section 2.4.

2.2.1 Axisymmetric model

In the axisymmetric model, we assume that the shape of an air shower is cylindric, with its symmetry axis along the moving shower core. This model is valid for very distant showers, where the radius of the shower front near the ground does not change a lot with respect to the shower depth. There are two effects which cause asymmetry, i.e. which cause a deviation from the naively expected equal-signal circles. On the one hand, the footprint of an inclined shower, i.e. a cylindric section, is no longer a circle, but an ellipse. However, this “asymmetry” is a pure geometric effect. It can be calculated analytically, so that, when projecting the signal into the shower plane, trivially via trigonometric expressions, this asymmetry vanishes. The equations for the projections, see Figure 2.2.1, are given through

$$\ell = \rho \sin \theta, \quad (2.3)$$

$$r = \rho \cos \theta. \quad (2.4)$$

On the other hand, the attenuation effect (see Section 2.3.2) causes asymmetry in the signal, since downstream particles have to travel a larger distance in the atmosphere so that they are more attenuated than the upstream particles. This effect is the only real asymmetry-causing phenomenon in the axisymmetric model, since it does not vanish when projecting signal into the shower plane.

2.2. GEOMETRICAL MODEL OF AN EXTENSIVE AIR SHOWER

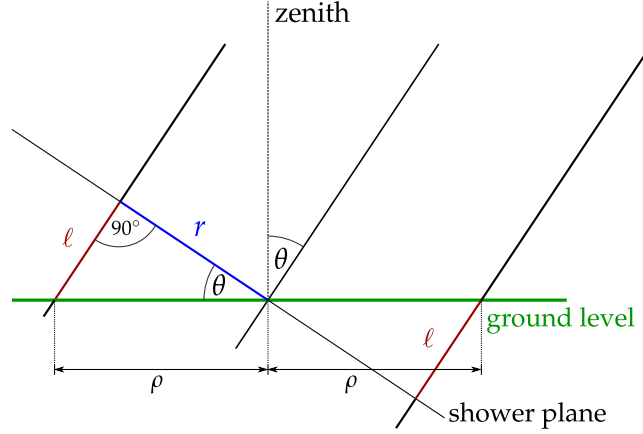


Figure 2.2.1: Sketch of a shower in the axisymmetric model including the definitions of the zenith angle θ , the radius of the shower in the shower plane r and in the ground plane ρ . It is important to note, that ρ is not a constant but a function of the azimuth angle in the ground plane.

2.2.2 Conical model

Now, the particles are assumed to be emitted from a single point with a specific opening angle α relative to the shower axis. This model is still simple but more realistic as we are modeling the air-shower development as a cone, whose apex coincides with the particle-emission point. The geometric parameters (see again Figure 2.1.1 and Figure 2.2.2), which are necessary for the later calculations, are $\vec{L}$, the vector between the conical apex at $d \hat{a}$ and the station position $\vec{\rho}$, and the angle α , under which, from the apex, station is seen relative to the shower core. With distance d from the shower core to the apex, the vector $\vec{L}$ is

$$\vec{L} = d \hat{a} - \vec{\rho} \quad \text{with} \quad L^2 = r^2 + (d - r \tan \theta \cos \psi)^2. \quad (2.5)$$

From Figure 2.2.2 we see that the opening angle α can be expressed as

$$\sin \alpha = \frac{r}{L} = \frac{r}{\sqrt{r^2 + (d - r \tan \theta \cos \psi)^2}}. \quad (2.6)$$

We give one remark to this equation. If r is fixed, $\sin \alpha$ will change with ψ so that the maximum of $\sin \alpha$ (and thus α) is reached at the upstream point ($\psi = 0^\circ$) and its minimum at the downstream point ($\psi = \pm 180^\circ$). This means that, for exactly the same r , downstream particles are emitted from the apex at smaller angles α , than upstream particles.

Contrarily, for a constant apex angle α under which particles are emitted, we obtain for the radius in the shower plane

$$r = \frac{d}{\tan \theta \cos \psi + d/r_y}, \quad (2.7)$$

where r_y is the radius in the neutral direction, $\psi = \pm 90^\circ$. This means that a conical shower produces constant α contours with shower-plane radius r as a function of the shower-plane angle ψ .

Additionally, we define another angle θ_L , the local zenith angle. It is the angle between a normal $\hat{n}$ at the station position and $\vec{L}$, the vector linking the apex (shower origin) and the impact point (see Figure 2.2.3). The local zenith angle is thus

$$\cos \theta_L = [\hat{n} \cdot \vec{L}] = \frac{d \cos \theta}{L} = \frac{d \cos \theta}{\sqrt{r^2 + (d - r \tan \theta \cos \psi)^2}}. \quad (2.8)$$

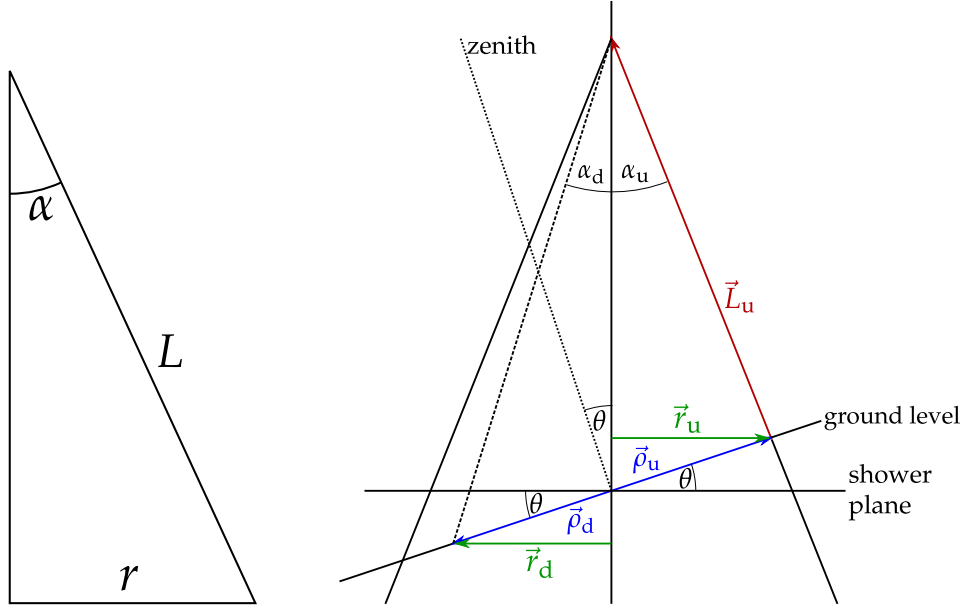


Figure 2.2.2: *Left:* Definition of the apex angle α . *Right:* Sketch of a shower in the conical model, including the definitions of the zenith angle θ , the distance from the core to the station in the shower plane $\vec{r}$ and in the ground plane $\vec{\rho}$, the distance vector $\vec{L}$ from the station to the shower origin, and the opening angle α at the apex. The indices u and d correspond to the upstream and the downstream direction. Note that the ground in this reference coordinate system is tilted while the arrival direction of the shower is always vertical. We see that $\alpha_d \neq \alpha_u$ even though $|\vec{r}_d| = |\vec{r}_u|$.

Note that the minimum value we can get for L is $L_{\min} = d \cos \theta$. Therefore, $\cos \theta_L \leq 1$ and a solution for θ_L can be always found.

2.3 Asymmetry causing effects

Before parameterizing the signal produced by an EAS, we first consider the two leading reasons for asymmetry in this signal, the geometrical effect of the detector shape and the attenuation effect.

2.3.1 Geometrical effect of the detector shape

At first we consider the geometrical shape of the SD stations. In the conical model all particles originate from the apex. Coming from this specific point, the particle impinges upon a SD station under zenith angle θ_L (see Figure 2.2.3). As a consequence, the cross-section area of the detector, as seen from the apex, varies as a function of θ_L (see Figure 2.3.1). This effective area can be formulated as shown in [19],

$$A_{\text{eff}} = A_{\text{top}} \cos \theta_L + A_{\text{side}} \sin \theta_L \equiv A_t(\theta_L), \quad (2.9)$$

where the two areas A_{top} and A_{side} represent the top area and the effective side area of a SD station. Since the detectors have a cylindrical shape, the areas are $A_{\text{top}} = \pi r_t^2$ and $A_{\text{side}} = 2 r_t h_t$, where r_t is the radius and h_t the height of a SD station. Consequently, the effective area is

$$A_t(\theta_L) = \pi r_t^2 \cos \theta_L + 2 r_t h_t \sin \theta_L. \quad (2.10)$$

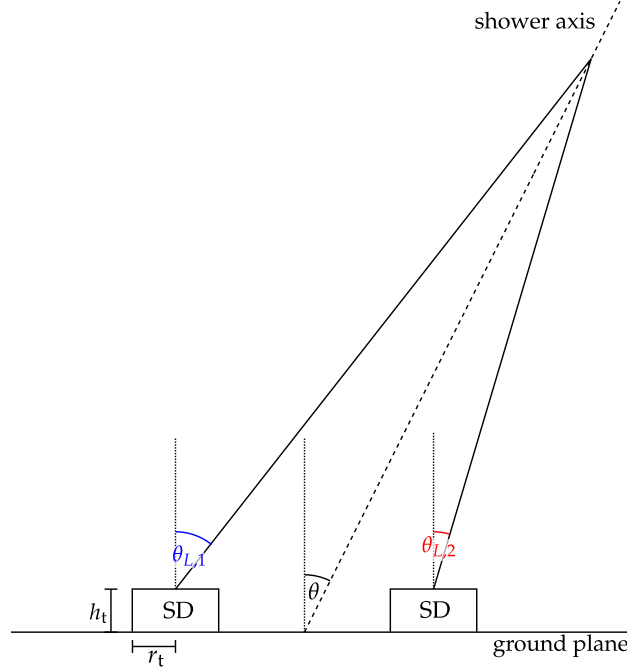


Figure 2.2.3: Illustration of the changing local zenith angle θ_L . One can see that θ_L at a downstream station is larger than at an upstream station.

A dependency of A_t on θ_L is shown in Figure 2.3.1. We see that, up to a maximum at around 23° , the effective area increases from $\sim 10.2 \text{ m}^2$ to $\sim 11.1 \text{ m}^2$. After the maximum, the curve decreases rapidly down to $\sim 7.5 \text{ m}^2$ at $\theta_L = 70^\circ$. This means, since the detected signal is proportional to $A_t(\theta_L)$, the more inclined the shower, the lower the signal. Furthermore, at a constant radius r , the angle θ_L is smaller at the upstream than at the downstream side. So, at least for angles larger than 23° , the effective area of the detector is larger at the upstream side. This means that an asymmetry in measured signal arises, which causes higher signal in the upstream than in the downstream direction.

2.3.2 Attenuation of charged particles

Secondly, we examine the attenuation effect. Irrespective of the underlying geometric model, for an inclined shower, $\theta \neq 0$, particles in the downstream part of the shower always cover a larger distance to reach the ground than particles in the upstream part. Since the density of particles decreases exponentially with the growing travel distance, the downstream part of the shower will be more attenuated than the upstream part. Consequently, the measured signal at downstream side will decrease and asymmetry arises in upstream direction. This effect is negligible for the hadronic shower component, because the hadrons hit the ground narrowly around the shower axis. It thus has to be considered only for the electromagnetic and the muonic part. However, muons interact much less than electrons. Therefore, the attenuation asymmetry reaches its maximum in the electromagnetic shower.

For a covered distance L , the attenuation effect can be written as

$$f_{\text{att}}(L) = \exp\left(-\frac{L}{\lambda}\right). \quad (2.11)$$

Here, λ is the attenuation length, which varies for different particles like electrons, positrons, or muons. For muons, for example, it typically ranges from 29 to 36 km [1].

Note that in the conical model the general length L is the absolute value of $\vec{L}$ (see Section 2.2.2).

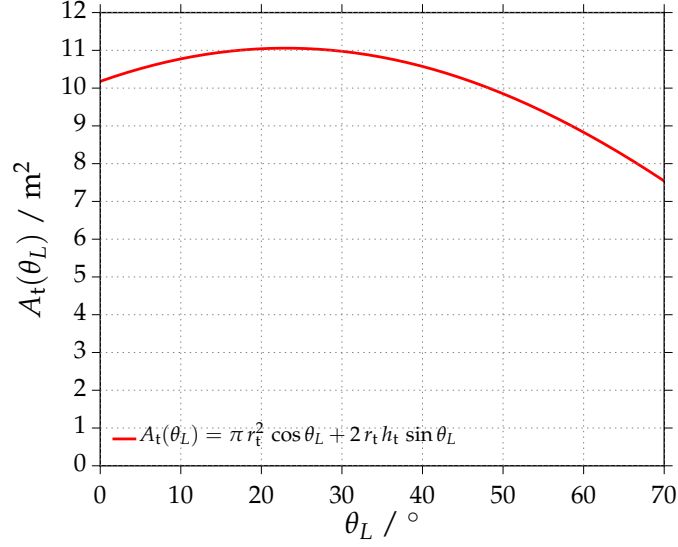


Figure 2.3.1: Plot of the effective detector area $A_t(\theta_L)$, as seen from the apex, with the parameters $r_t = 1.8 \text{ m}$ and $h_t = 1.2 \text{ m}$ for increasing local zenith angle θ_L .

2.4 Signal at the ground

The task is now to parametrize the signal S , produced by a shower at the ground. It is important to repeat that in the following, calculations are made in the conical model only.

We start with the fact that the number of particles N that hit the ground is proportional to

$$N \propto \text{ADF}(\alpha) \Delta\Omega. \quad (2.12)$$

Here, $\text{ADF}(\alpha) = dN/d\Omega$ is the angular distribution function of the emission of particles from the apex. $\Delta\Omega$ is the solid angle under which the detector is seen from the apex, which is

$$\Delta\Omega = A_t(\theta_L)/L^2, \quad (2.13)$$

where $A_t(\theta_L)$ is the effective detector area. There is a difference of the deposited signal in the water-Cherenkov stations between muons and electrons. The electrons only penetrate few centimeters into the water. Thus the deposited signal is proportional to the effective detector area $A_t(\theta_L)$ as seen from the actual zenith angle. In contrast, the signal of a single muon is proportional to the length of its trajectory inside the water. The longer the path in the SD station, the larger is the gained signal. The number of received muons is proportional to the effective area of the detector. As a consequence, the the signal proportionality turns into

$$S = \text{ADF}_\mu(\alpha) \frac{A_t(\theta_L) \bar{\ell}(\theta_L)}{L^2 \ell_{\text{scale}}} k_\mu + \text{ADF}_{\text{em}}(\alpha) \frac{A_t(\theta_L)}{L^2} k_{\text{em}} \quad (2.14)$$

$$= \frac{1}{L^2} \left(\text{ADF}_\mu(\alpha) \frac{V_t}{\ell_{\text{scale}}} k_\mu + \text{ADF}_{\text{em}}(\alpha) A_t(\theta_L) k_{\text{em}} \right), \quad (2.15)$$

where $\bar{\ell}(\theta_L)$ represents the mean trajectory length of the muons and ℓ_{scale} corresponds to a length scale. Furthermore, $A_t(\theta_L) \bar{\ell}(\theta_L)$ is equal to the volume $V_t = A_t h_t$ of the water-Cherenkov tank. By choosing $\ell_{\text{scale}} = h_t$, the height of a station, the signal is measured in units of VEM (vertical equivalent muon) and the factor k_μ becomes simply $k_\mu = 1 \text{ VEM}$ by definition. Finally, with $V_t/h_t = A_t = \pi r_t^2 = \text{const.}$, we obtain

$$S = \frac{1}{L^2} \left[\text{ADF}_\mu(\alpha) A_t k_\mu + \text{ADF}_{\text{em}}(\alpha) A_t(\theta_L) k_{\text{em}} \right]. \quad (2.16)$$

2.5. DISTANCE TO THE SHOWER MAXIMUM

Additionally, we insert the attenuation factor $f_{\text{att}}(L)$, as given in Section 2.3.2. In conclusion, the final signal is given by

$$S = \frac{1}{L^2} \left[\text{ADF}_\mu(\alpha) A_t k_\mu + \text{ADF}_{\text{em}}(\alpha) A_t(\theta_L) k_{\text{em}} \right] f_{\text{att}}(L). \quad (2.17)$$

Now, we simplify the ADF-term. Although the electromagnetic and the muonic terms are producing slightly different zenith-angle dependencies, we will model the signal with a single ADF. Since the muonic dominates the electromagnetic signal, $k_{\text{em}} \ll k_\mu$ [6, 20], we obtain

$$\text{ADF}_\mu(\alpha) A_t k_\mu + \text{ADF}_{\text{em}}(\alpha) A_t(\theta_L) k_{\text{em}} \approx k \text{ADF}(\alpha) A_t, \quad (2.18)$$

where k is close to 1 VEM. The angular distribution function itself we will model as a power law,

$$\text{ADF}(\alpha) = \kappa \left(\frac{\alpha}{\alpha_0} \right)^{-\gamma}. \quad (2.19)$$

The attenuation factor in the conical model turns into $f_{\text{att}}(L) = \exp(-L/\lambda)$ where $L = |\vec{L}|$ as given in Equation (2.11). The final signal model is

$$S = k \text{ADF}(\alpha) \frac{A_t}{L^2} \exp\left(-\frac{L}{\lambda}\right) \quad (2.20)$$

$$\propto \frac{1}{L^2} \exp\left(-\frac{L}{\lambda}\right) \alpha^{-\gamma}. \quad (2.21)$$

At the end of this Chapter, we give an illustration of the behaviour of the so far derived variables α and r in the shower as well as in the ground plane (Figure 2.4.1). For a fixed set of d , θ and r_y , the contours in the x-y-plane are shown, where either α or r is constant. Additionally, the contours of equal signal are given. They are calculated with (2.21).

2.5 Distance to the shower maximum

So far, we derived a model for the signal an EAS produces on the ground. Inside this model the parameter L occurs as a function of r , d , θ , and ψ . However, for a (simulated) EAS we do not know the distance d to the shower apex, but only the depth of the shower maximum X_{max} , which is obtained by a fit of the longitudinal shower profile to the Gaisser-Hillas function [18]. This motivates the following Section, in which we will derive an expression for d with respect to the zenith angle θ and the depth of the apex X_{ap} , which is chosen to follow the shower evolution as a certain fraction of X_{max} .

As a simplification, the atmosphere is assumed to be isothermal and thus exponential. At first, for a vertical shower, the shower depth as a function of the height h is given by

$$X(h) = \int_h^\infty \rho(h') dh' = X_0 \exp\left(-\frac{h}{h_{\text{scale}}}\right), \quad (2.22)$$

where $\rho(h)$ is the atmospheric density at a height h , X_0 the depth of the ground level (for the Pierre Auger Observatory about 860 g/cm^2), and h_{scale} the effective scale of the (isothermal) atmosphere. For a vertical shower, the height h equals the distance d , but for an inclined shower the height increase with distance is slower, $h(d) = d \cos \theta$ (see Figure 2.5.1, left). The depth at a certain distance d from the shower core at the ground is thus

$$\begin{aligned} X(d) &= \int_d^\infty \rho(d) dd = \int_{h(d)}^\infty \rho(h(d)) \frac{dh}{\cos \theta} \\ &= \frac{1}{\cos \theta} \int_h^\infty \rho(h) dh = \frac{1}{\cos \theta} X_0 \exp\left(-\frac{h(d)}{h_{\text{scale}}}\right) \\ &= \frac{1}{\cos \theta} X_0 \exp\left(-\frac{d \cos \theta}{h_{\text{scale}}}\right). \end{aligned} \quad (2.23)$$

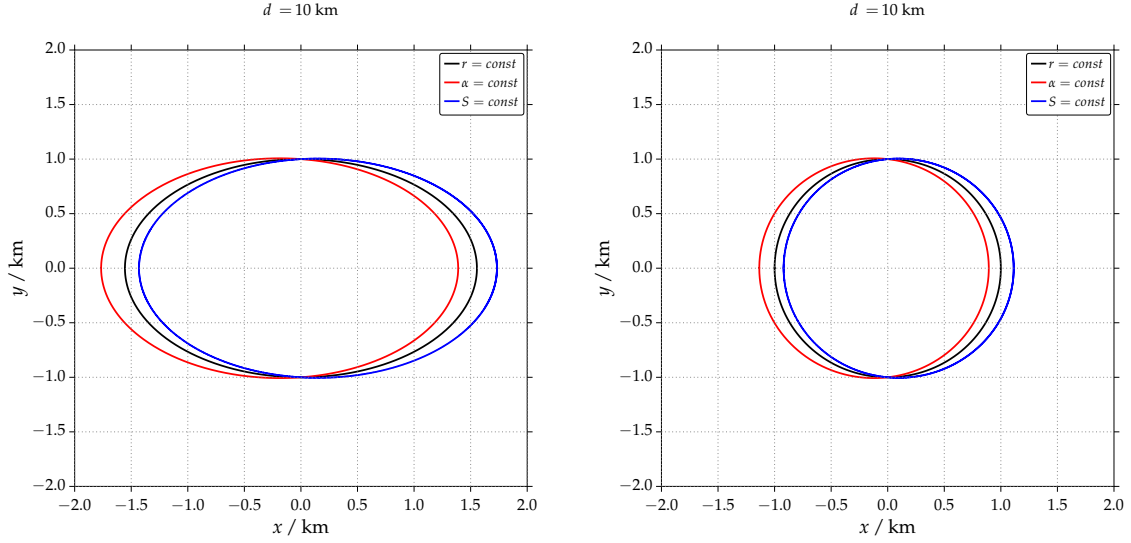


Figure 2.4.1: Contours of constant radius r (black), constant apex angle α (red), and constant signal S (blue) shown in the ground plane (left) and in the shower plane (right). The used parameters are the zenith angle $\theta = 50^\circ$, apex distance $d = 10$ km, “neutral” $r_y = 1000$ m, attenuation length $\lambda = 36$ km, and power-law exponent $\gamma = 1.4$ (see Section 2.3.2 and 2.4 for their definition). A positive x -axis is in upstream direction and a negative x -axis is in downstream direction. For more examples of such contours, see Appendix D.

To find the distance d to the apex we set this depth to the depth of the apex, $X(d) = X_{\text{ap}}$. Now the distance d to the apex can be expressed as

$$\begin{aligned} d(\theta) &= -\frac{h_{\text{scale}}}{\cos \theta} \ln \left(\frac{X_{\text{ap}}}{X_0} \cos \theta \right) \\ &= h_{\text{scale}} \sec \theta \ln \left(\frac{X_0 \sec \theta}{X_{\text{ap}}} \right). \end{aligned} \quad (2.24)$$

2.6 Expansion of the signal

Finally, we will expand the derived signal formula (2.21). A proper expansion parameter for the signal is the factor r/d , since we obtain $r/d \ll 1$ for distant showers, $X_0 \gg X_{\text{ap}}$. The factor r/d we shortly call ε . It is implicit in the variables L and thus in α ,

$$\begin{aligned} L &= \sqrt{r^2 + (d - r \tan \theta \cos \psi)^2} = d \sqrt{\left(\frac{r}{d}\right)^2 + \left(1 - \frac{r}{d} \tan \theta \cos \psi\right)^2} \\ &= d \sqrt{\varepsilon^2 + (1 - \varepsilon \tan \theta \cos \psi)^2}, \end{aligned} \quad (2.25)$$

$$\sin \alpha = \frac{r}{L} = \frac{\varepsilon}{\sqrt{\varepsilon^2 + (1 - \varepsilon \tan \theta \cos \psi)^2}}. \quad (2.26)$$

The goal is, to get a linearized form of the signal. Therefore, we will successively expand all factors of the signal in first order of ε .

Primarily, we expand $\sin \alpha$. By using the relations $\sin \alpha \approx \alpha$ for small angles and $x/(1 - ax) \approx x(1 + ax)$ we obtain for α

$$\alpha \stackrel{O(\varepsilon)}{\approx} \varepsilon (1 + \varepsilon \cos \psi \tan \theta). \quad (2.27)$$

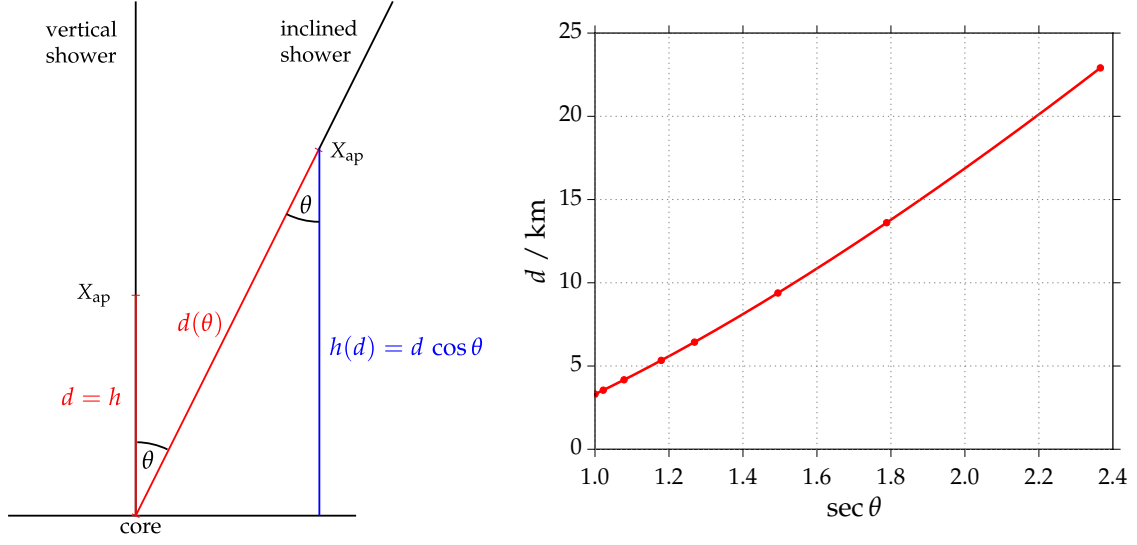


Figure 2.5.1: The picture on the *left* shows a construction sketch for derivation of the apex distance $d(\theta)$ in equation (2.24). The relation between the apex distance d and the depth of the apex X_{ap} as well as the dependency on zenith angle θ is given on the *right*, where the behavior of d versus $\sec \theta$ is shown for the parameters $X_0 = 860 \text{ g/cm}^2$, $X_{\text{ap}} = 550 \text{ g/cm}^2$, and $h_{\text{scale}} = 7.4 \text{ km}$. The points are given for the zenith angles $\theta = 0^\circ, 12^\circ, 22^\circ, 32^\circ, 38^\circ, 48^\circ, 56^\circ$, and 65° .

Subsequently, we insert this result into the power law, so that the ADF turns into

$$\begin{aligned} \text{ADF}(\alpha) &= \kappa \left(\frac{\varepsilon}{\alpha_0} \right)^{-\gamma} (1 + \varepsilon \cos \psi \tan \theta)^{-\gamma} \\ &\approx \kappa \left(\frac{\varepsilon}{\alpha_0} \right)^{-\gamma} (1 - \gamma \varepsilon \cos \psi \tan \theta) \\ &\propto (1 - \gamma \varepsilon \cos \psi \tan \theta). \end{aligned} \quad (2.28)$$

The second factor of equation (2.21), arisen from the solid angle, turns into

$$\frac{1}{L^2} = \frac{1}{d^2 [\varepsilon^2 + (1 - \varepsilon \cos \psi \tan \theta)^2]} \stackrel{O(\varepsilon)}{\approx} \frac{1}{d^2} (1 + 2 \varepsilon \cos \psi \tan \theta). \quad (2.29)$$

Finally, we expand the function $f_{\text{att}}(L)$, describing the attenuation effect, and we obtain

$$f_{\text{att}}(L) = \exp \left(-\frac{d}{\lambda} \sqrt{\varepsilon^2 + (1 - \varepsilon \cos \psi \tan \theta)^2} \right) \stackrel{O(\varepsilon)}{\approx} \exp \left(-\frac{d}{\lambda} \right) \left(1 + \frac{d}{\lambda} \varepsilon \cos \psi \tan \theta \right). \quad (2.30)$$

Concluding, we insert the expansions of the three factors into equation (2.21) and then we are left with

$$S(r, \psi) \propto \frac{1}{d^2} \exp \left(-\frac{d}{\lambda} \right) \left[1 + \left(2 - \gamma + \frac{d}{\lambda} \right) \varepsilon \cos \psi \tan \theta + O(\varepsilon^2) \right]. \quad (2.31)$$

The goal was to linearize the signal and we arrived at the following shape

$$S(r, \psi) = S(r) (1 + a \cos \psi), \quad (2.32)$$

where a is the asymmetry parameter containing

$$a = \left(2 - \gamma + \frac{d(\theta)}{\lambda} \right) \frac{r}{d(\theta)} \tan \theta. \quad (2.33)$$

For fitting and plotting purposes the asymmetry parameter

$$b = \frac{a}{\tan \theta}, \quad (2.34)$$

so that

$$S(r, \psi) = S(r) (1 + b \tan \theta \cos \psi), \quad (2.35)$$

turns out to be more suitable. With this at least a part of the θ -dependency is extracted from the asymmetry parameter. For this reason, and as a motivation for the following Chapter, not the expanded a is shown but b (Figure 2.6.1, left). Additionally, getting the magnitude of b from the exact formula (2.21) in two extreme directions $\psi = 0^\circ$ and $\psi = 180^\circ$ relative to the neutral direction at $\psi = \pm 90^\circ$, we get

$$\bar{b}_{\text{exact,up}} = \left(\frac{S(r, 0^\circ)}{S(r, 90^\circ)} - 1 \right) \frac{1}{\tan \theta}, \quad (2.36)$$

$$\bar{b}_{\text{exact,down}} = \left(1 - \frac{S(r, 180^\circ)}{S(r, 90^\circ)} \right) \frac{1}{\tan \theta}, \quad (2.37)$$

which is also shown. Note here that the x-axis is using $\sin^2 \theta$, which from natural distribution of measured data later turns out to be a more suitable variable. The result for one set of fixed values is shown in Figure 2.6.1 (left).

In this plot one can see that both curves, which originate from the non-expanded formula, have a similar shape. At first, they increase steeply up to a certain maximum in the $\sin^2 \theta$ -range of 0.1 to 0.2. Afterwards, the curves decrease less steeply down to a much smaller value than at their beginning. It is conspicuous that the curves for up- and downstream are not exactly the same. They differ at some points in more than 0.05, which is $\sim 20\%$. This is caused by the fact that, for a fixed radius r , there is no pure $\cos \psi$ -dependency of the signal at small zenith angles. Although the equal signal line is still a circle, the shower core is not exactly at the center of this circle. The core is shifted in the upstream direction.

The curve of the expanded formula (grey curve in Figure 2.6.1, left) lies in-between the upstream and the downstream function, at least in the θ -range of $\sin^2 \theta > 0.4$. This means that the asymmetry extracted in the linearized form describes the mean asymmetry. Subsequently, the expansion to the first order of ε is justified for this zenith angle range.

However, it is striking that the curve, originating from the expanded formula, does not match the two other curves at all in the lower θ area. This is based on the θ -dependency of d (see Section 2.5). For a low zenith angle θ , also the distance d is small when X_{ap} is close to X_0 . Consequently, for small zenith angle θ , the expansion assuming $r/d \ll 1$ is not valid anymore. Figure 2.6.1 (right) shows ε as a function of $\sin^2 \theta$. It is obvious that in the lower $\sin^2 \theta$ -range $\varepsilon \ll 1$ is no longer valid.

Finally, there is a third remark regarding Figure 2.6.1, which is important to mention. It is about the plot parameter γ , which is the exponent of the term forming the $\text{ADF}(\alpha)$. Since $\text{ADF}(\alpha)$ is an empirical function, also γ is empirical. In this plot, it is chosen to be 1.4, because the resulting plot matches best with the results in the analysis part.

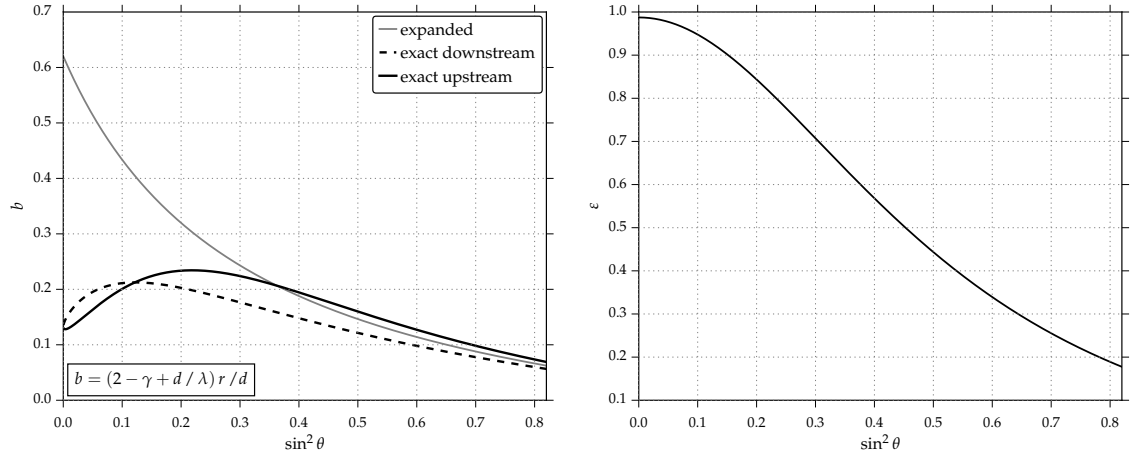


Figure 2.6.1: *Left:* Plot of the asymmetry parameter b with respect to $\sin^2 \theta$ for fixed values of $X_0 = 860 \text{ g/cm}^2$, $X_{\text{ap}} = 750 \text{ g/cm}^2$, $h_{\text{scale}} = 7.4 \text{ km}$, $\gamma = 1.4$, $\lambda = 36 \text{ km}$, and $r = 1 \text{ km}$. The zenith angle θ ranges from 0° to 65° . Three curves are shown representing the function for upstream (*black*) and downstream (*dashed*) direction (see Equations (2.36) and (2.37)) as well as the function arising from the expanded formula (2.34) (*grey*). *Right:* Size of the expansion parameter ε with respect to $\sin^2 \theta$.

Chapter 3

Analysis

After the discussion of theoretical aspects of the signal asymmetry, arising for SD measurements, in the previous Chapter, the analysis of data is the issue now. We analyze how the “real” data behave and whether the derived expansion of the signal and its asymmetry is appropriate or not. The term “real” in the prior sentence is quoted because the analysis is done with simulation data, not with real measured data. This is motivated by the fact that, for the simulation data, important parameters like primary energy and primary mass are known. Only with the knowledge of these parameters, it will be possible to analyze and subsequently parametrize the asymmetry, which could be used in future improvements of the SD reconstructions.

For this analysis, the data library, called NAPOLI, is used which contains signal data for different interaction models, primaries, energy bins, and zenith angles. This library is *continuous* in energy as well as in zenith angle θ . The energy range starts at $E = 10^{18}$ eV and ends at $E = 10^{20}$ eV, the zenith angle gets values from 0° to 65° . Out of this data, for all possible combinations of the parameters, the signal is fitted with $S_a(\psi) = S_n (1 + a \cos \psi)$ as well as $S_b(\psi) = S_n (1 + b \tan \theta \cos \psi)$, where a and b are the asymmetry parameters and S_n is the signal in the “neutral” stations, i.e. at $\psi = \pm 90^\circ$. Afterwards, the dependence of the asymmetry parameters a and b on the different parameters (interaction model, zenith angle, primary mass, and primary energy) is analyzed. The goal is to find regularities in these dependencies which form a global parametrization of a or b . The fitting with the formula above already implies a linearization of the signal with respect to $\cos \psi$. Therefore, this thesis also aims to examine whether this linearization is valid or not.

Note here that the analysis is exclusively done for dense ring stations, i.e. special stations located in simulation at a constant radius of $r = 1000$ m. Therefore, for this analysis, among all other parameters, it was not necessary to bin different radii, which reduces the dimensionality of the problem. However, the behaviour of the asymmetry in r would be another interesting item to analyze.

Before considering the asymmetry, the first subsection gives some motivation and explanations of how the data were actually evaluated.

3.1 Analysis settings

3.1.1 Binning of zenith angle and energy

The NAPOLI library contains data with simulation parameters structured in a tree-diagram way. This means that, as first, we distinguish between two interaction models, QGSJET and EPOS. The second level of the “tree” is the differentiation between the four primaries, proton (p), helium (He), oxygen (O), and iron (Fe). Then, for each of the four primaries, every single simulated event is saved. One single event consists of a set of parameters: energy E , depth of the shower maximum $X_{\max}$, and zenith angle θ , and for each set of parameters a sublist of the station parameters is given, for those stations which received some signal. More explicitly, this sublist of station parameters

contains the station ID, the radius r , the shower-plane azimuth angle ψ , and the signal S . Since the analysis in this thesis focuses on dense ring stations for SD measurement, the important values are only ψ and S . A dense ring is a special set of 12 stations available only in simulations and placed on a ring with exactly 1000 m distance from the shower core in the shower-plane projection, i.e. $r = 1000 \text{ m} = \text{const.}$.

The NAPOLI library is continuous, i.e. the energy as well as the zenith angle θ are not available in discrete values but vary from event to event within the whole available range. Therefore, it is necessary to bin both of these variables to be able to fit functions to the data. Here, two conditions have to be taken into consideration. On the one hand, the bin width has to be large enough in order to get satisfying statistics in each bin. On the other hand, the bin width must not be too large because the goal is to analyze the fit parameters in dependence of e.g. θ and energy. If the bins were too large, one could not observe the dependencies precisely. Only very general and global behaviours could be found out.

Hence, histograms are produced for the different variables in order to illustrate the reason for the choice of the bin shape. In Figure 3.1.1 the θ -binning is shown. Since in the Monte Carlo simulation the zenith angle θ was generated from a uniform distribution in $\sin^2 \theta$, we use this sinusoidal bin shape. The uniform distribution in $\sin^2 \theta$ is motivated by the natural distribution of the zenith angle, which also shows this dependency.¹

What also is chosen out of the histograms is the bin number. It is set to the value of 50. For this number, the counts are within the range of 2000 to 3000 which is large enough for satisfying statistics. Figure 3.1.2 shows histograms of the energy dependent on the primary and the interacting model. These histograms suggest a linear binning in $\lg(E)$. The bin width is chosen to be 0.2 so that there arises ten equal shaped energy-bins.

3.1.2 Profile plots

The signal S of the stations is plotted against the azimuth angle ψ (see Figure 3.1.3, left). Since ψ represents the location of the stations (upstream, downstream), its range starts at -180° and ends at 180° where those angles are the same. For the dense ring, there are exactly 12 stations which measure the signal. So, ψ is by construction binned in 12 equally shaped bins with a bin width of 30° . Finally, for each combination of the model, primary particle type, energy bin, and θ bin, all S - ψ -pairs are collected and plotted in one graph. Additionally, each data point has a certain error in S , which is also considered. The two functions $S_a(\psi) = S_n(1 + a \cos \psi)$ as well as $S_b(\psi) = S_n(1 + b \tan \theta \cos \psi)$ with the asymmetry parameters a and b are fitted to those data sets via the `scipy.optimize.curve_fit` package in *python* [21]. However, for reasons of clarity, not the plots including all data points are shown but only the so called profile plots. These plots show the mean of each ψ -bin and the fluctuation in each bin as error bars. In Figure 3.1.3 the comparison between an all-data points-plot and a profile plot is shown.

Two remarks have to be given here. First, in the later analysis, only the asymmetry factor b is used for further evaluations, not a anymore. The extraction of the initial θ -dependency (at small θ) out of the asymmetry parameter is the reason for favoring b . It is possible to extract this θ -dependency because the conical model (Section 2.2.2) implies the $\tan \theta$ term. The second remark is about the dense ring stations. The SD reaches its full detection efficiency only for energies larger than 3 EeV. This means that, for events with energies $\lg(E / \text{eV}) \in [18.1, 18.3, 18.5]$, it is still possible that some of the 12 stations do not have triggers. This causes a bias of the data towards larger signals in the lower energy range. When analyzing the asymmetry in the following sections, this effect will be addressed again.

3.1.3 Fit parameters

In this section, we show how the fit parameters a, b , and S_n are computed. The `scipy.optimize.curve_fit` function calculates this automatically, so that the values need not be calculated manually. How-

¹ The natural distribution consists of product of a $\cos \theta$ -term, which originates from the plane detector area, and a $\sin \theta$ -term, which originates from the solid angle under which the sky is seen. This leads to a uniform distribution in $\sin^2 \theta$.

3.1. ANALYSIS SETTINGS

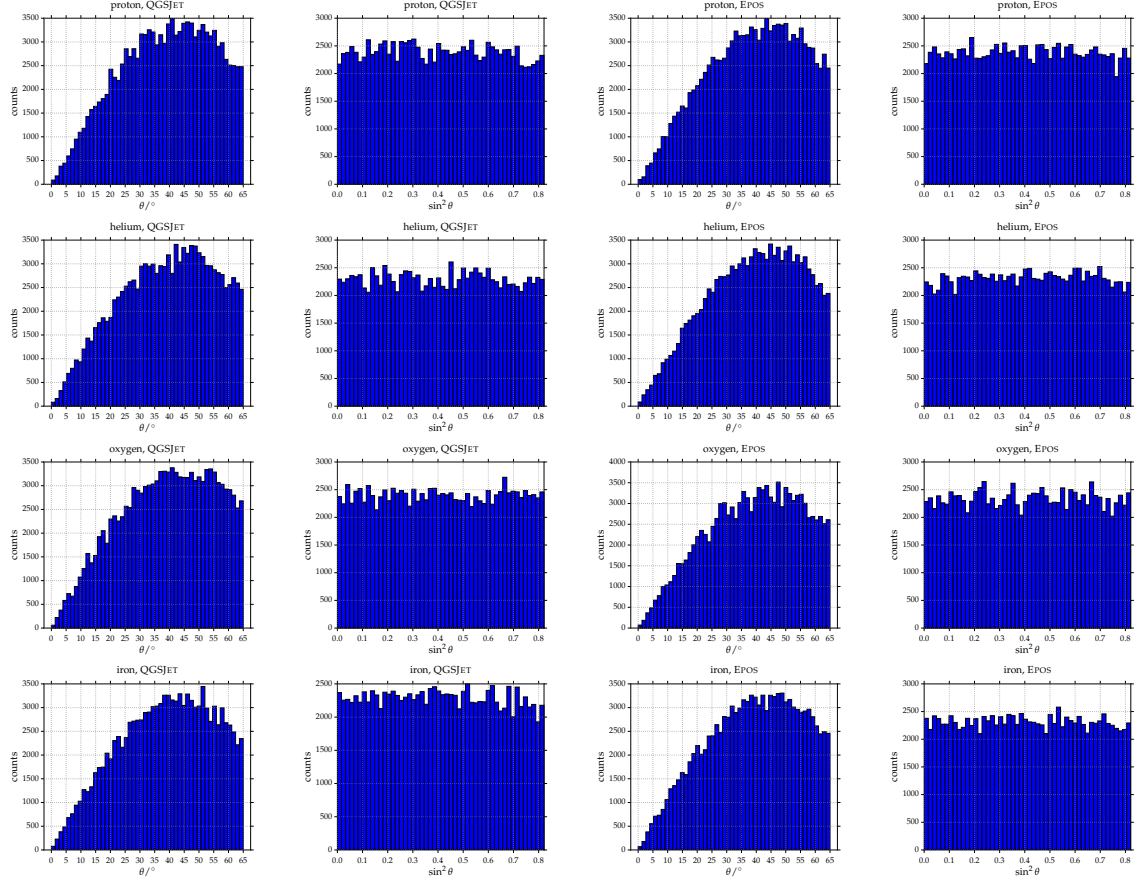


Figure 3.1.1: Motivation for the zenith-binning. The figure shows histograms of θ for different primaries in the two simulation models QGSJET and EPOS for binning in 50 equally shaped θ -bins as well as for binning in 50 equally shaped $\sin^2 \theta$ -bins. The four rows correspond to proton, helium, oxygen, and iron. The four columns represent the binning in θ and $\sin^2 \theta$ for QGSJET followed by the same binning for EPOS.

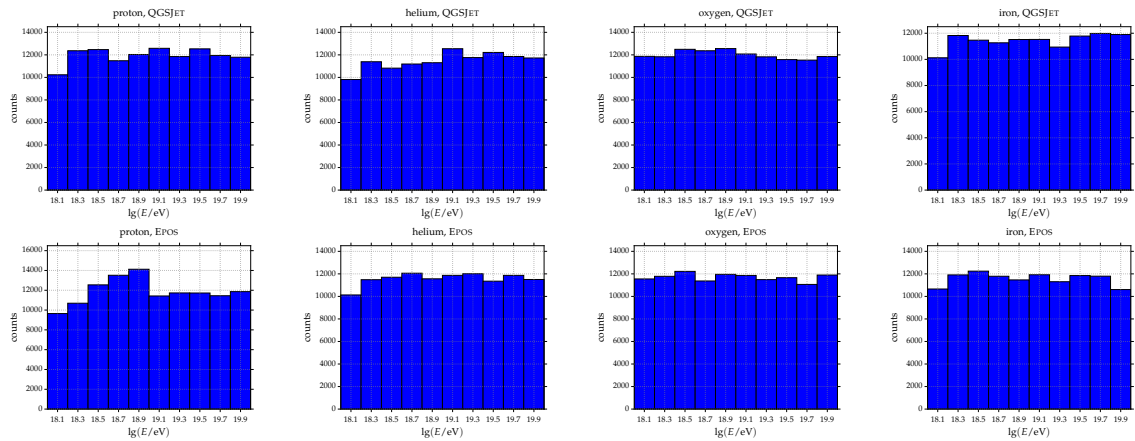


Figure 3.1.2: Motivation for the energy-binning. The figure shows histograms of the primary particle energy for different primary types in the two simulation models QGSJET and EPOS for binning in 10 equal energy-bins. The two rows correspond to the two interacting models, the four columns represent the primaries proton, helium, oxygen, and iron.

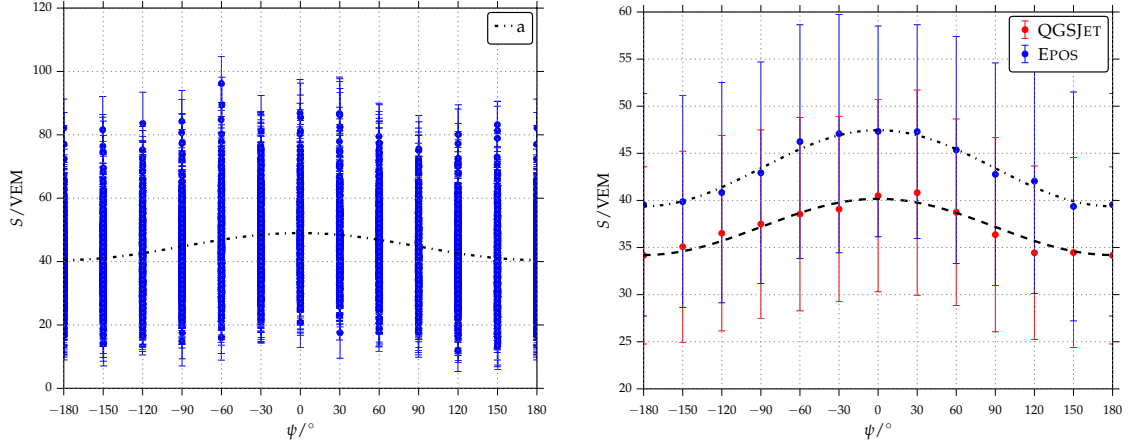


Figure 3.1.3: Motivation for profile plots. *Left:* Plot of all signals for zenith angle $\theta = 65^\circ$, energy $\lg(E / \text{eV}) = 19.5$, proton as the primary particle, and the QGSJET model. *Right (blue dots):* The same data set shown as a profile plot.

ever, to illustrate the procedure, the calculation is outlined below. Note that here the fit functions $S_a(\psi) = S_n (1 + a \cos \psi)$ and $S_b(\psi) = S_n (1 + b \tan \theta \cos \psi)$ are reformulated into

$$S(\psi) = S_n + A \cos \psi, \quad (3.1)$$

where $A = S_n a$ and $A = S_n b \tan \theta$, respectively.

Beginning with the least-squares method for each of the 12 dense-ring stations, where S_i is the measured signal value, we have to minimize

$$\chi^2 = \sum_{j=1}^{12} \chi_j^2(S_n, A) = \sum_{j=1}^{12} \sum_i^{\text{events}} [S_i - S(S_n, A)]_j^2. \quad (3.2)$$

From the requirement for the minimum two equations arise,

$$\sum_{j=1}^{12} \frac{\partial \chi_j^2}{\partial S_n} \equiv 0 \quad \text{and} \quad \sum_{j=1}^{12} \frac{\partial \chi_j^2}{\partial A} \equiv 0. \quad (3.3)$$

When inserting $S(S_n, A) = S_n + A \cos \psi$ into equations (3.3), the following formula results,

$$\begin{aligned} \sum_{j=1}^{12} \frac{\partial \chi_j^2}{\partial S_n} &= \sum_{j=1}^{12} \sum_i^{\text{events}} \frac{\partial}{\partial S_n} (S_{ij} - S_n - A \cos \psi_{ij})^2 = \sum_{k=1}^N -2 (S_k - S_n - A \cos \psi_k) \equiv 0 \\ &\Rightarrow \sum_k S_k - N S_n - A \sum_k \cos \psi_k = 0, \end{aligned} \quad (3.4)$$

$$\begin{aligned} \sum_{j=1}^{12} \frac{\partial \chi_j^2}{\partial A} &= \sum_{j=1}^{12} \sum_i^{\text{events}} \frac{\partial}{\partial A} (S_{ij} - S_n - A \cos \psi_{ij})^2 = \sum_{k=1}^N 2 \cos \psi_k (S_k - S_n - A \cos \psi_k) \equiv 0 \\ &\Rightarrow \sum_k S_k \cos \psi_k - S_n \sum_k \cos \psi_k - A \sum_k \cos^2 \psi_k = 0. \end{aligned} \quad (3.5)$$

For reasons of abbreviating we introduce the notations $\sum_k x_k / N \equiv \bar{x}$, $\sum_k (x_k y_k) / N \equiv \overline{xy}$, and $\sum_k \cos \psi_k / N \equiv \bar{C}$. We obtain the following system of equations,

$$\bar{S} - S_n - A \bar{C} = 0, \quad (3.6)$$

$$\overline{SC} - S_n \bar{C} - A \overline{C^2} = 0. \quad (3.7)$$

This system of equations has the solution

$$A = \frac{\bar{S}\bar{C} - \overline{SC}}{\bar{C}^2 - \overline{C^2}}, \quad (3.8)$$

$$S_n = \bar{S} - A\bar{C} = \bar{S} - \frac{\bar{S}\bar{C} - \overline{SC}}{\bar{C}^2 - \overline{C^2}}\bar{C}. \quad (3.9)$$

With this background information about the analysis settings, the real analysis can be started. Since the hadronic interaction model as well as the primary mass are expected to be less important than the dependency on θ and energy, they are analyzed separately first. In contrast to that, the θ and energy dependencies are not that easy to separate. That is why they are considered in the same section.

3.2 Dependency on hadronic interaction model

The first dependency which is analyzed is the impact of the underlying hadronic interaction model of the shower simulation on the signal and its asymmetry. There exist two different models in the NAPOLI library which are used for shower simulations, the Epos-LHC [22, 23] and the QGSJET-II.04 [24–26] model. Both interaction models are post-LHC models, i.e. both models are tuned with data from the LHC at CERN. The most important difference between them is the total number of muons, which is higher in the Epos than in the QGSJET model. This fact implies at least one expectation: the mean of the signal in each ψ bin should be always larger in the Epos than in the QGSJET model out of the fact that there is a higher absolute number of particles which produce signal.

Out of the fits to the NAPOLI data (as described in Section 3.1) the asymmetry parameters a and b are computed and stored for each combination of the model, primary particle type, energy, and zenith angle. Figure 3.2.1 shows an example plot which is created out of these asymmetry parameters. In order to see the dependency on the interacting models, both curves, the one for the QGSJET and the one for the Epos model, are shown in the same plot. In Chapter A of the appendix, the other resulting plots for each bin combination are attached. It turns out that both curves have a very similar shape. That means that the dependency on the interacting model is not dominant. Nonetheless, a slight discrepancy between the two curves in the area for $\theta > 40^\circ$ occurs. The asymmetry parameters in the Epos model are $<3\%$ larger than in the QGSJET model.

For parameterizing the asymmetry, this slight discrepancy does not play any crucial role. Therefore, we will not distinguish between the two models when fitting an empirical function to the data. The points from QGSJET and Epos we will both use for the same fit.

3.3 Dependency on primary mass

In the NAPOLI library there are four different primary particles (mass number A) for which simulations have been produced: protons (p, $A = 1$), helium (He, $A = 4$), oxygen (O, $A = 16$), and iron (Fe, $A = 56$). These four different data sets are shown together in one plot (see Figure 3.3.1 and Appendix B). The data points of p and Fe are highlighted in terms of color because they are the upper and the lower limit of each θ -bin, respectively. This shows that the asymmetry changes in the same direction for increasing or decreasing primary mass approximately linearly in logarithmic mass, $\ln(A)$. However, there is in fact a difference between the asymmetry factors of the four primaries, which appears roughly for $\theta \gtrsim 35^\circ$. Here the curves spread up to a range of about 5 to 10 %. The heaviest nucleus exhibits the lowest asymmetry and vice versa. The value of 5 to 10 % means that the impact of the primary mass on the asymmetry of the signal S is twice as large as the impact of the interaction model. Considering the global behaviour of the curves, though, the primary mass is, like the interaction model, not the dominating factor. Therefore, similarly to the hadronic interaction model, we will not differentiate between the four different

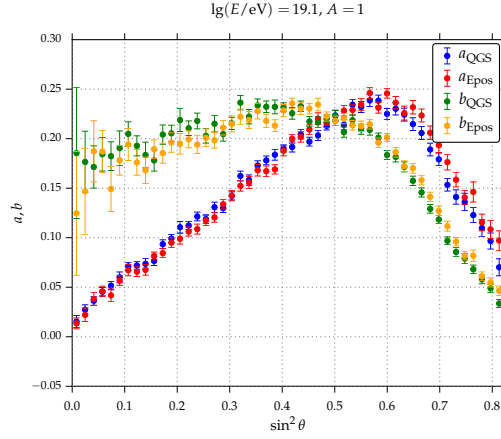


Figure 3.2.1: Example plot of the asymmetry parameters a and b versus $\sin^2 \theta$ comparing the two different interaction models QGSJET and Epos, for the energy of $\lg(E / \text{eV}) = 19.1$ and proton as the primary particle.

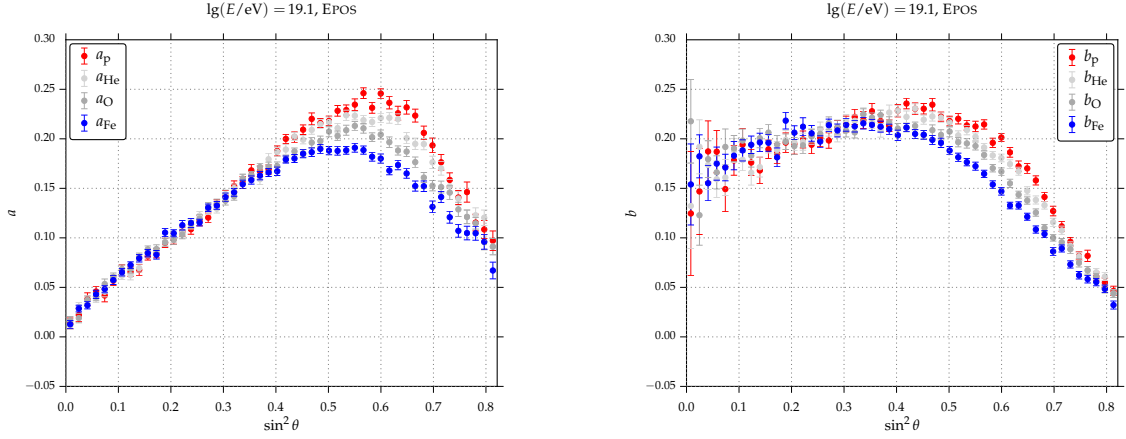


Figure 3.3.1: Example plot of the asymmetry parameters a (left), b (right) versus $\sin^2 \theta$ comparing the four different primaries p, He, O and Fe for the Epos model and within the energy bin of $\lg(E / \text{eV}) = 19.1$.

primary masses later, when fitting a function to the data.

Consequently, the linearized asymmetry lost two degrees of freedom, the interaction model and the primary mass. This makes it much easier to fit an empirical function to the data in order to identify the other remaining dependencies. These empirical fits are explained in the following Section.

3.4 Dependency on zenith angle and energy

3.4.1 Fit setup

As it is described in Section 3.2 and 3.3, the asymmetry has now two parameters left, the primary energy and the zenith angle θ . The leading θ -dependency can be extracted with the $\tan \theta$ term. What remains are the data sets for b , which are shown in the Appendix B. An empirical function

has to be found which describes the data sufficiently. We have modeled the data with

$$b = \underbrace{(k_0 + k_1 \sin^2 \theta)}_{\text{linear term}} \underbrace{\frac{1}{1 + \exp(-(\sin^2 \theta - x_0) / x_1)}}_{\text{logistic term}}. \quad (3.10)$$

This is a function consisting of a linear part for small values of $\sin^2 \theta$ and a logistic part for large values of $\sin^2 \theta$, with the four free parameters k_0 , k_1 , x_0 , and x_1 . Here, k_0 represents the offset of the linear part, k_1 is the slope. In addition, x_0 equals to the $\sin^2 \theta$ -value of the cusp of the logistic part, whereas x_1 is the slope in this point. For shortening, the notations

$$x \equiv \sin^2 \theta, \\ \text{logistic}(x; x_0, x_1) \equiv \frac{1}{1 + \exp(-(x - x_0)/x_1)},$$

are introduced. Hence, equation (3.10) turns into a compact form

$$b = (k_0 + k_1 x) \text{logistic}(x; x_0, x_1). \quad (3.11)$$

This empirical function is now fitted to the data, i.e. for each of the ten energy bins, where all data of model and primaries are included, the four parameters are computed again with the `scipy.optimize.curve_fit` package in *python* [21]. The fits are shown in Figure 3.4.1. The energy dependence of the fit parameters is illustrated in Figure 3.4.2 (left). Before interpreting the results, a slight modification of the fit function is done. Considering the blue curve in Figure 3.4.2 (left), a clear dependence on energy is visible. The blue dots represent the parameter k_0 , which is the offset of the linear area of the b -curves in the first $\sin^2 \theta$ -bin. In order to get a more energy independent parameter, the function is shifted in that way, that the new offset represents the y -value of the b -curve in that $\sin^2 \theta$ -bin, where it is most independent of the energy. For the data, this is the 18th $\sin^2 \theta$ -bin, which means $\sin^2 \theta \approx 0.3039 \equiv x_k$. Consequently, the fit function turns into

$$b = [k_0 + k_1 (x - x_k)] \text{logistic}(x; x_0, x_1), \quad (3.12)$$

$$\equiv (k_2 + k_1 x) \text{logistic}(x; x_0, x_1), \quad (3.13)$$

with $k_2 \equiv k_0 - k_1 x_k$ as the modified offset parameter.

3.4.2 Results

The result of the fits, which are described in the previous Section, is shown in Figure 3.4.2 (right), the values are given in Table 3.1. With these fits, the energy dependence is separated from the θ -dependency. The four fit parameters contain the whole energy variability. Therefore, they are analyzed more detailed in the following.

x_0 . The parameter x_0 (red dots in Figure 3.4.2) represents the cusp point in the logistic part of the fit. It shows an increasing shape in the lower energy part up to $\lg(E / \text{eV})$ between 18.5 and 18.7. From this energy on it is constant. The first interpretation would than be that x_0 possesses an energy dependency in the lower energy range. However, the SD of the Pierre Auger Observatory is at these energies not fully efficient yet. For energies smaller than 3 EeV ($\lg(E / \text{eV}) \approx 18.47$), not necessarily all of the twelve dense-ring stations obtain signal. There is a threshold at a certain signal level, which cuts off some of the stations. This selection effect causes the descent in the x_0 curve at lower energies. To get the value of x_0 for each energy, one would extrapolate the constant curve in the higher energy range to the lower values. Consequently, x_0 is not energy dependent but an empirical constant.

x_1 . The slope of the logistic function in the cusp point is described by the variable x_1 , which is represented by the yellow curve in Figure 3.4.2. For the whole energy range, it is constant around -0.1 . Therefore, x_1 is not energy dependent, too, like x_0 . Thus, it appears that the logistic part of the fit function is quite universal.

k_2 . In contrast to the logistic part, the linear part shows a non-constant behaviour in energy. k_2 , the b value of the $\sin^2 \theta$ -bin for which b is most constant in energy ($\sin^2 \theta \approx 0.3039$), shows a decrease of about 0.06 after being constant up to $\lg(E / \text{eV}) = 19.3$.

k_1 . Moreover, the slope of the linear part of the fit function k_1 (green curve in 3.4.2) varies with increasing energy even more. It fluctuates within a range from 0.1 to 0.3 in absolute value. In the lower energy part, it has lower values. From $\lg(E / \text{eV}) = 18.7$ onwards, it increases again.

In summary, the empirical function of the asymmetry parameter b can be parametrized as

$$b = [k_2(E) + k_1(E) x] \text{ logistic}(x; x_0, x_1), \quad (3.14)$$

where the energy dependence and the θ -dependence are separated. This equation can be inserted in the linearized signal formula $S_b(\psi) = S_n (1 + b \tan \theta \cos \psi)$ and we obtain a parametrization of the signal, which depends on the zenith angle θ , the primary energy, and the azimuth angle ψ .

Additionally, it turns out, that the conical model (Sections 2.2, 2.4, and 2.6) developed in this thesis does not match the data well. This is caused by the fact that the parameters in the conical model (γ , λ , and d) still have some residual dependence on E and θ , which does not follow from the considerations in Chapter 2, but is implied in the empirical function of the asymmetry we found in this thesis. Nevertheless, the conical model was useful for extracting the $\tan \theta$ -dependency of the asymmetry. For further studies it would be interesting to analyze the development of the asymmetry function for more complex shower-shape models, especially in the form of the Gaisser-Hillas function, with the aim of finding the residual dependencies of the model parameters on the energy and the zenith angle analytically.

3.4. DEPENDENCY ON ZENITH ANGLE AND ENERGY

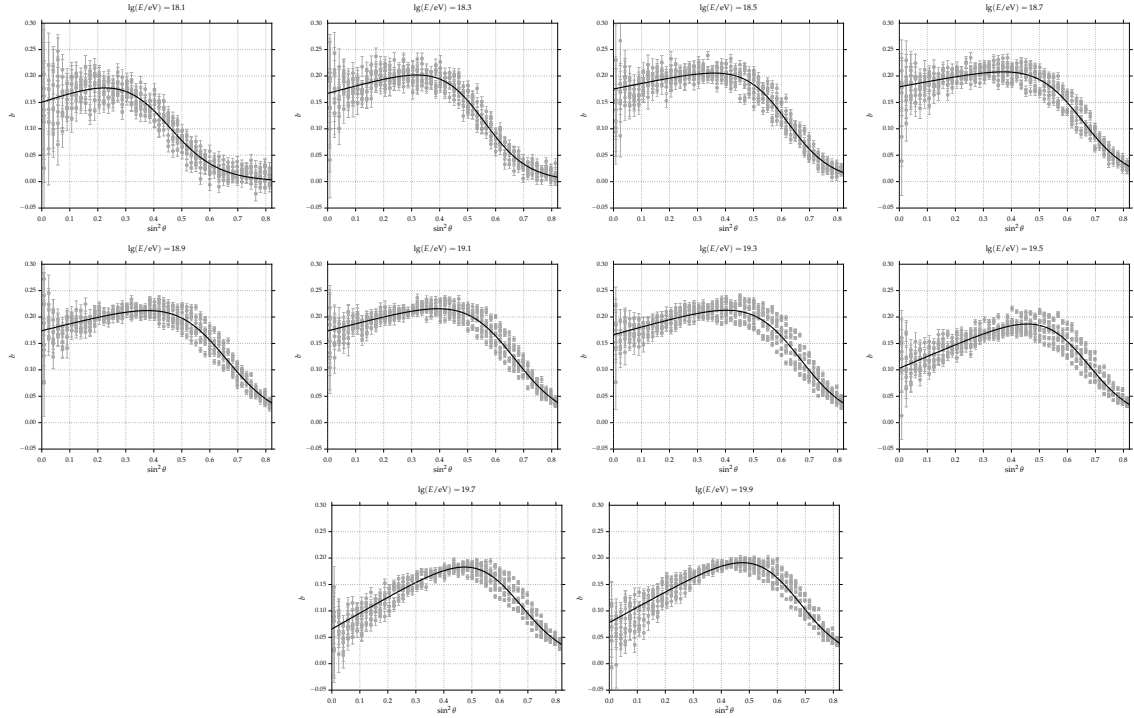


Figure 3.4.1: Empirical fit to the data sets, consisting of the two hadronic models and primary nuclei types in dependency on $\sin^2 \theta$. The grey dots and their uncertainties are all the data points from the four primary mass types and the two hadronic models, merged in a single data set. The empirical function is fitted (black curve) to all this data points at once, in contrast to Figure 3.2.1 and Figure 3.3.1, where it is differentiated between the interaction models as well as the primary masses. For these eight plots, from left to right and top to bottom, the energy bin increases.

Table 3.1: Values of the parameters of the empirical fits as well as their uncertainties for increasing energy bins.

$\lg(E / \text{eV})$	k_2	σ_{k_2}	k_1	σ_{k_1}	x_0	σ_{x_0}	x_1	σ_{x_1}
18.1	0.2086	0.0070	0.1885	0.0395	0.4330	0.0100	-0.0872	0.0027
18.3	0.2112	0.0020	0.1445	0.0156	0.5456	0.0043	-0.0791	0.0017
18.5	0.2082	0.0011	0.1075	0.0100	0.6105	0.0031	-0.0791	0.0014
18.7	0.2097	0.0009	0.0988	0.0084	0.6438	0.0029	-0.0853	0.0014
18.9	0.2150	0.0009	0.1340	0.0080	0.6433	0.0029	-0.0942	0.0013
19.1	0.2158	0.0007	0.1381	0.0066	0.6516	0.0022	-0.0906	0.0011
19.3	0.2114	0.0006	0.1468	0.0056	0.6511	0.0018	-0.0891	0.0009
19.5	0.1716	0.0005	0.2256	0.0050	0.6506	0.0017	-0.0853	0.0008
19.7	0.1579	0.0006	0.3034	0.0056	0.6428	0.0017	-0.0874	0.0008
19.9	0.1667	0.0005	0.2921	0.0053	0.6505	0.0016	-0.0869	0.0007

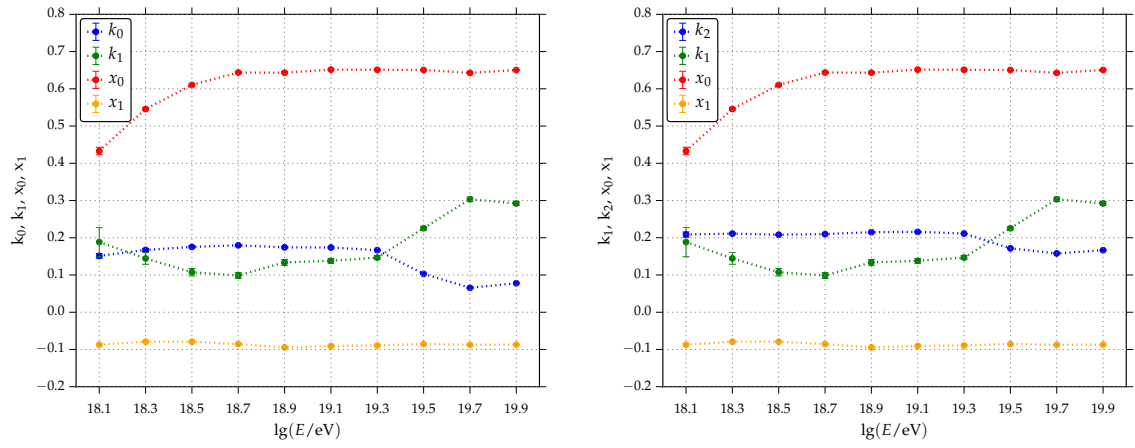


Figure 3.4.2: Energy dependence of the fit parameters k_0, k_1, x_0 , and x_1 (left) and k_1, k_2, x_0 , and x_1 (right) of the empirical fit to the NAPOLI data. The difference between the two plots is, that the parameter k_0 (left) is shifted so that it is more independent of energy. The shifted parameter is called k_2 (right).

Chapter 4

Conclusion and Prospects

This thesis pursued two goals. The first one was to develop a simplified theoretical model of air showers out of which it is possible to draw conclusions about the parametrization of the signal asymmetry in SD measurement at the Pierre Auger Observatory. The second goal was to find an empirical function for the signal asymmetry, which matches the simulated data.

In order to achieve the first aim, the very simplified model of a conical shower development was introduced with all of its necessary parameters. Then, the linearized form of the signal and its asymmetry was obtained, leading to the empirical formula

$$S(r, \psi) = S(r) (1 + b \tan \theta \cos \psi),$$

where b represents the asymmetry parameter and $S(r)$ describes the (mean) signal depending on the radius r from the shower core to the station. The asymmetry parameter b is in this simplified model given as

$$b = \left(2 - \gamma + \frac{d(\theta)}{\lambda} \right) \frac{r}{d(\theta)}.$$

where $d(\theta)$ is the distance between the shower core and the apex of the cone representing the shower development, λ is an attenuation length, and γ is the exponent of the angular distribution function, modeled as a power law. Subsequently, this model function was plotted for some fixed parameters as well as an expansion of the model in first order in r/d . An obvious difference between the upstream and the downstream asymmetry was the first result. Here, it became clear that, as a consequence of the asymmetry, a core shift appears when reconstructing the shower. Nevertheless, it emerged that the expansion of the formula is valid only in the high range of the parameter $\sin^2 \theta$. Moreover, the most important insight was to keep the explicit dependency of the asymmetry on $\tan \theta$. This was later used when fitting the simulation data.

After developing the model, the simulation data of the NAPOLI library were analysed. Since the data are continuous in energy and θ , they were binned in ten equally shaped bins of energy and 50 equal bins of $\sin^2 \theta$. Note here, that only dense ring stations with radius $r = 1000$ m were used for analysis, so that $S(r)$ became a constant. A further step would be to analyze the r dependency of the asymmetry. For that, not only the dense ring of stations should be used, but also all the other stations with varying r .

However, for the analysis of the dense rings only, the empirical function was developed in the form

$$b = [k_2(E) + k_1(E) x] \text{logistic}(x; x_0, x_1)$$

where $x \equiv \sin^2 \theta$ and $\text{logistic}(x; x_0, x_1)$ represents a logistic function with the break at x_0 and slope x_1 . The four parameters x_0, x_1, k_1 , and k_2 were obtained for the whole dataset and it turned out, that only the parameters of the linear function, k_1 and k_2 , are energy dependent.

CHAPTER 4. CONCLUSION AND PROSPECTS

Furthermore, it turns out that the asymmetry does not depend substantially on the particular choice of the interaction model, neither does it depend strongly on the primary mass. Therefore, a parametrization depending on the zenith angle θ and energy only is sufficient.

Besides the r -dependency, which was already mentioned before, two other issues arise from this analysis and should be considered. On the one hand, further studies about the relative magnitude of the different asymmetry-causing effects are of interest. On the other hand, a modification of the reconstruction of air showers with respect to the asymmetry is desirable. The parametrization of asymmetry developed in this thesis could be used to improve these reconstruction processes.

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Appendix

The Appendix contains figures which are mentioned in the text in various chapters, but are attached here for reasons of readability.

Appendix A

The following plots show the dependency of the asymmetry parameters a and b on $\sin^2 \theta$ as well as θ for increasing energy, with respect to the hadronic interaction models QGSJET and EPOS. The energy ranges from $\lg(E/\text{eV}) = 18.1$ to 19.9 in steps of 0.2 . The red (EPOS) and blue (QGSJET) dots represent the parameter a , the yellow (EPOS) and green (QGSJET) dots represent the parameter b . Such a collection of plots is shown for each of the four primaries, proton, helium, oxygen, and iron.

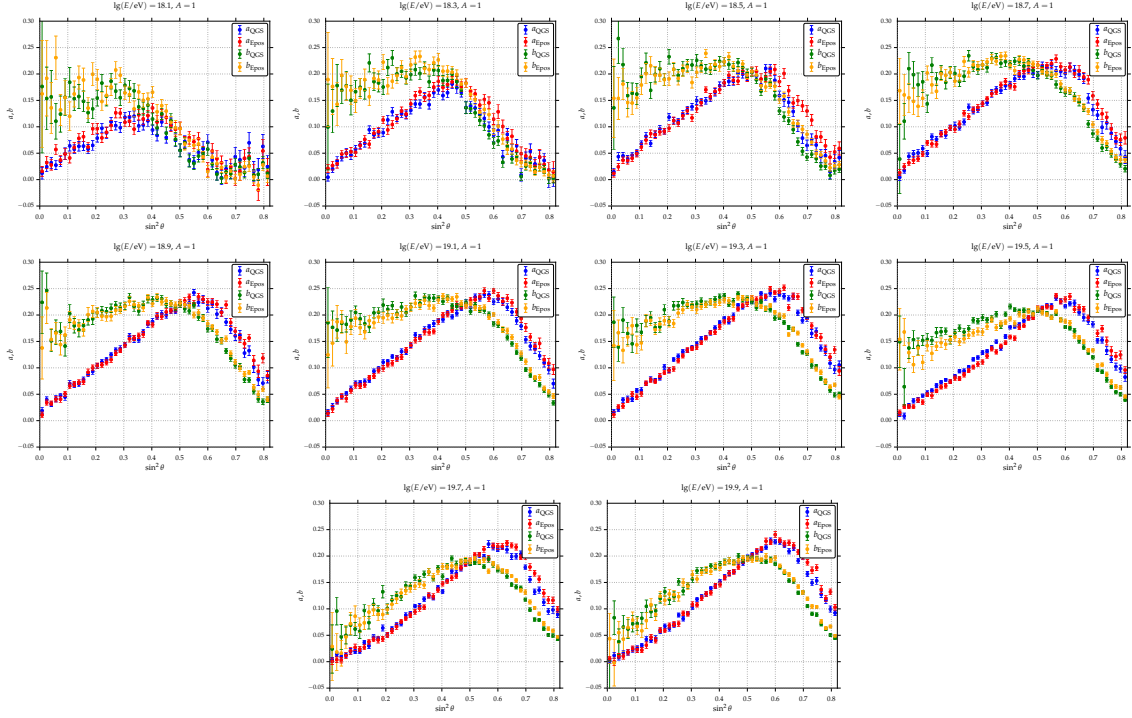


Figure 1: Plots of the asymmetry parameters a and b as a function of $\sin^2 \theta$ for proton as the primary particle and for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

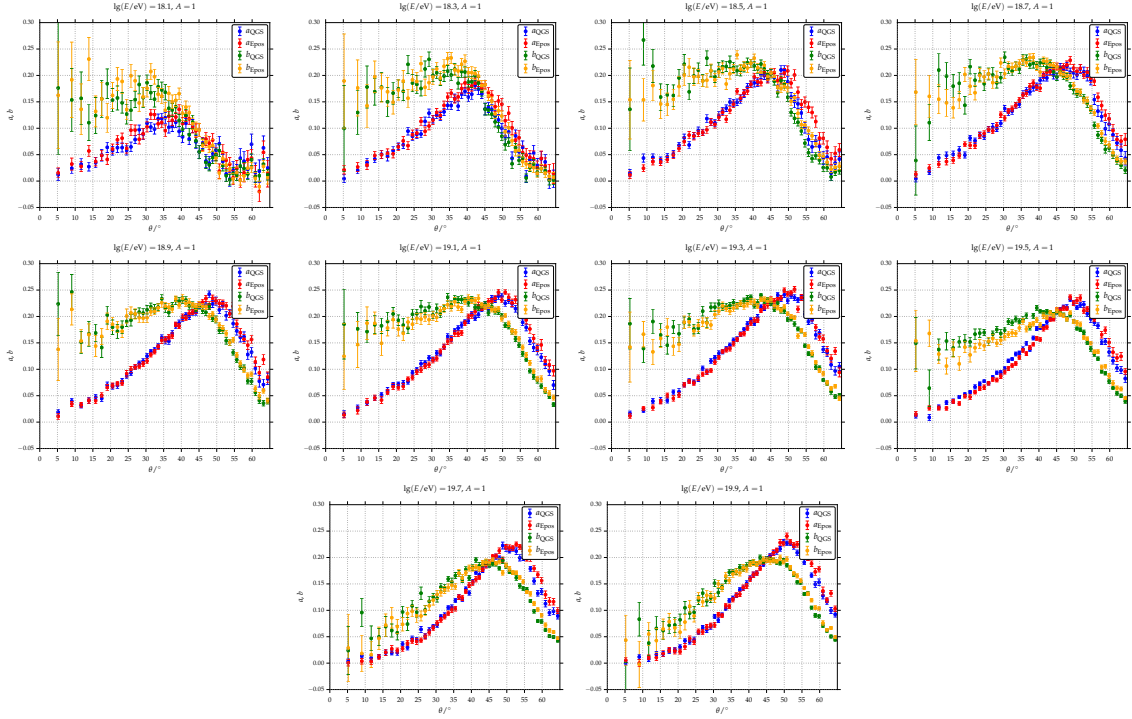


Figure 2: Plots of the asymmetry parameters a and b as a function of the zenith angle θ for proton as the primary particle and for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

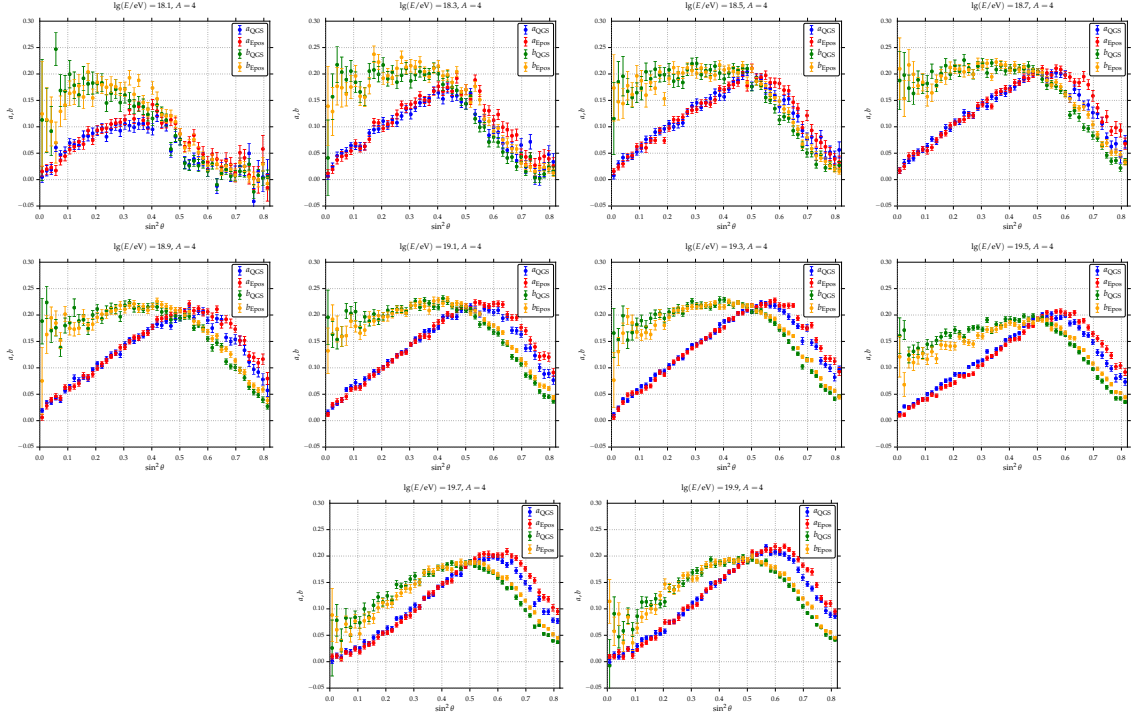


Figure 3: Plots of the asymmetry parameters a and b as a function of $\sin^2 \theta$ for helium as the primary particle and for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

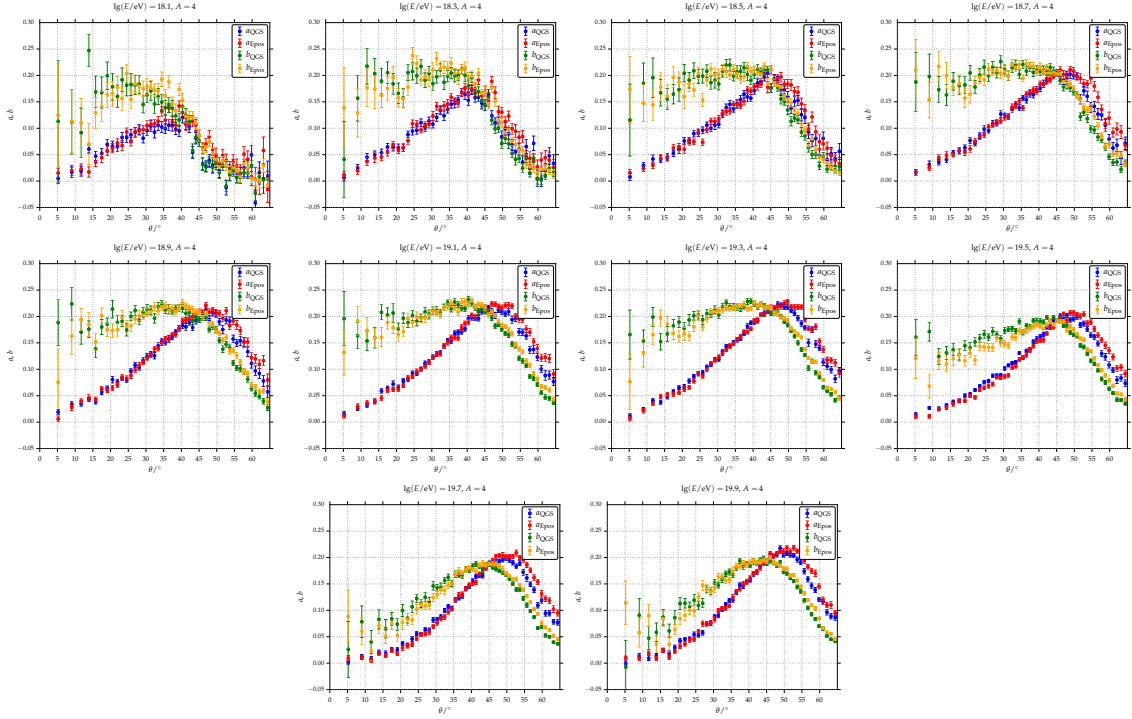


Figure 4: Plots of the asymmetry parameters a and b as a function of the zenith angle θ for helium as the primary particle and for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

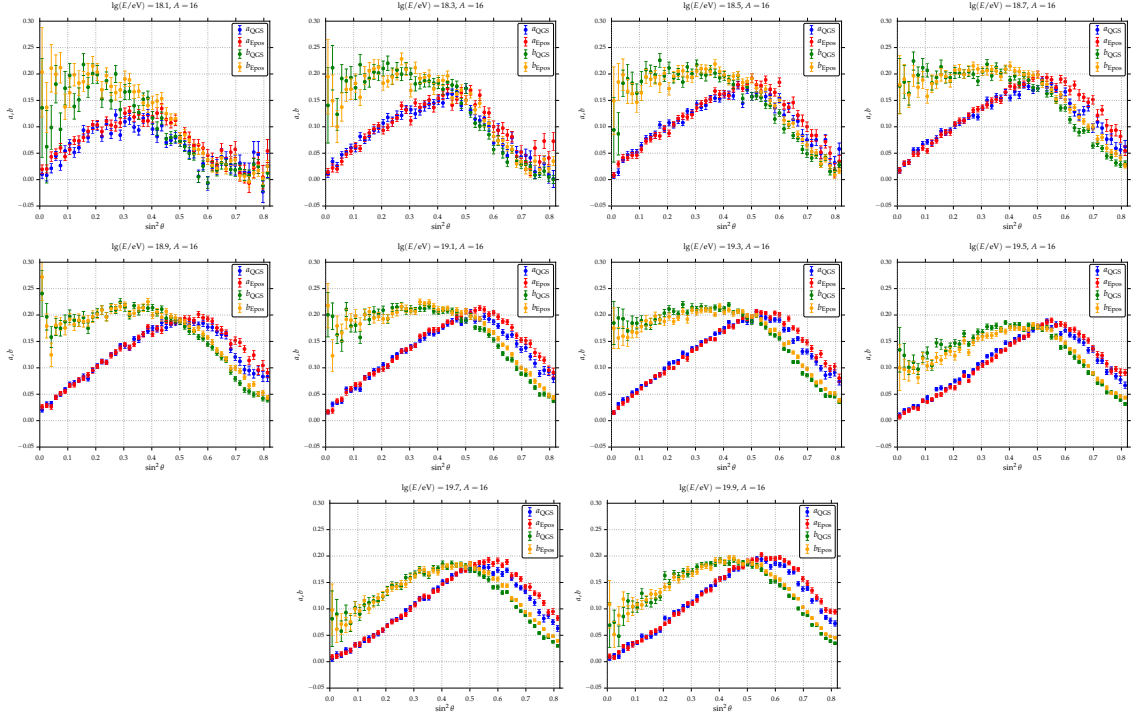


Figure 5: Plots of the asymmetry parameters a and b as a function of $\sin^2 \theta$ for oxygen as the primary particle and for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

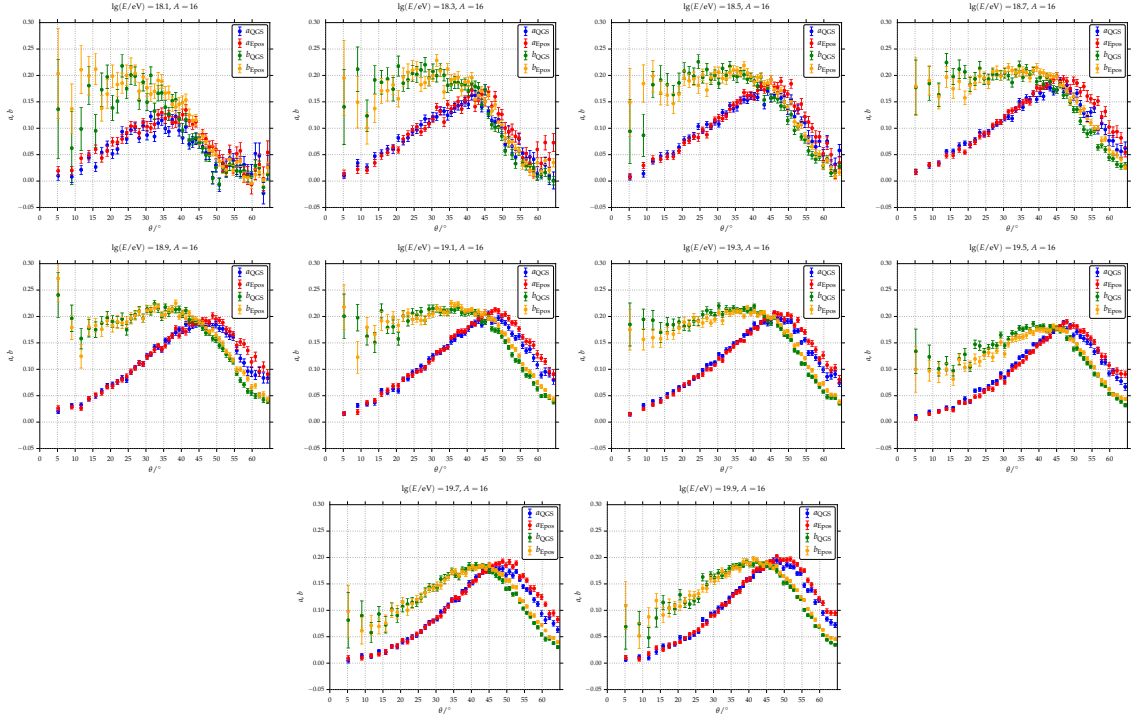


Figure 6: Plots of the asymmetry parameters a and b as a function of the zenith angle θ for oxygen as the primary particle and for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

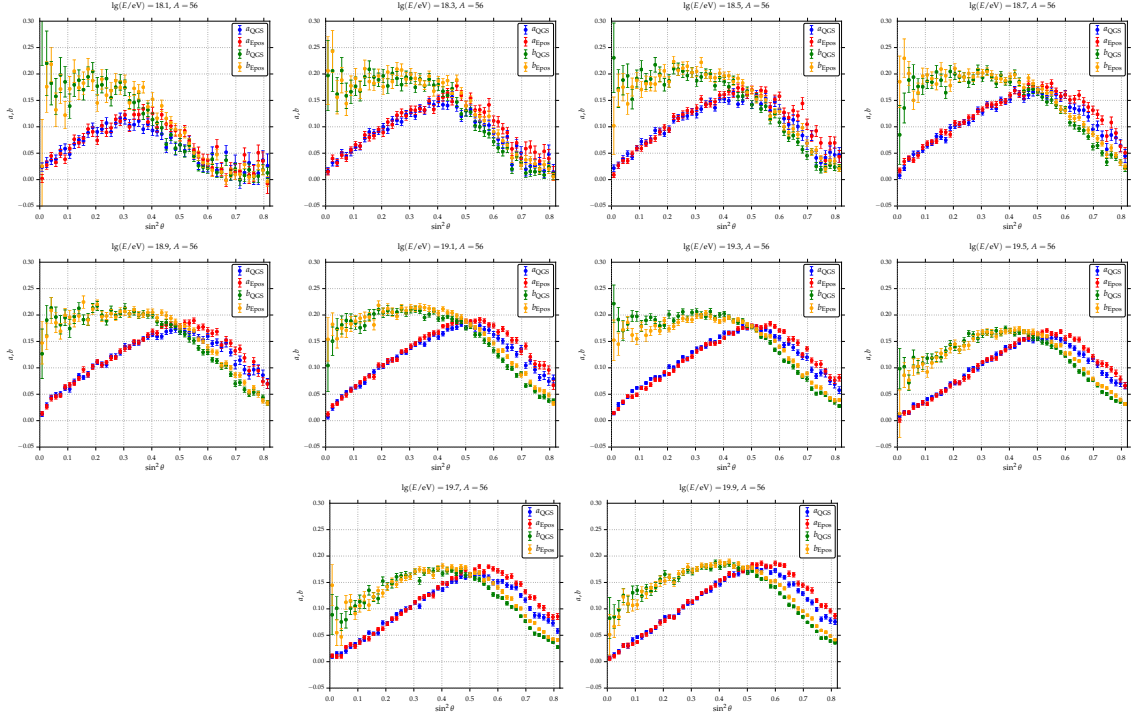


Figure 7: Plots of the asymmetry parameters a and b as a function of $\sin^2 \theta$ for iron as the primary particle and for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

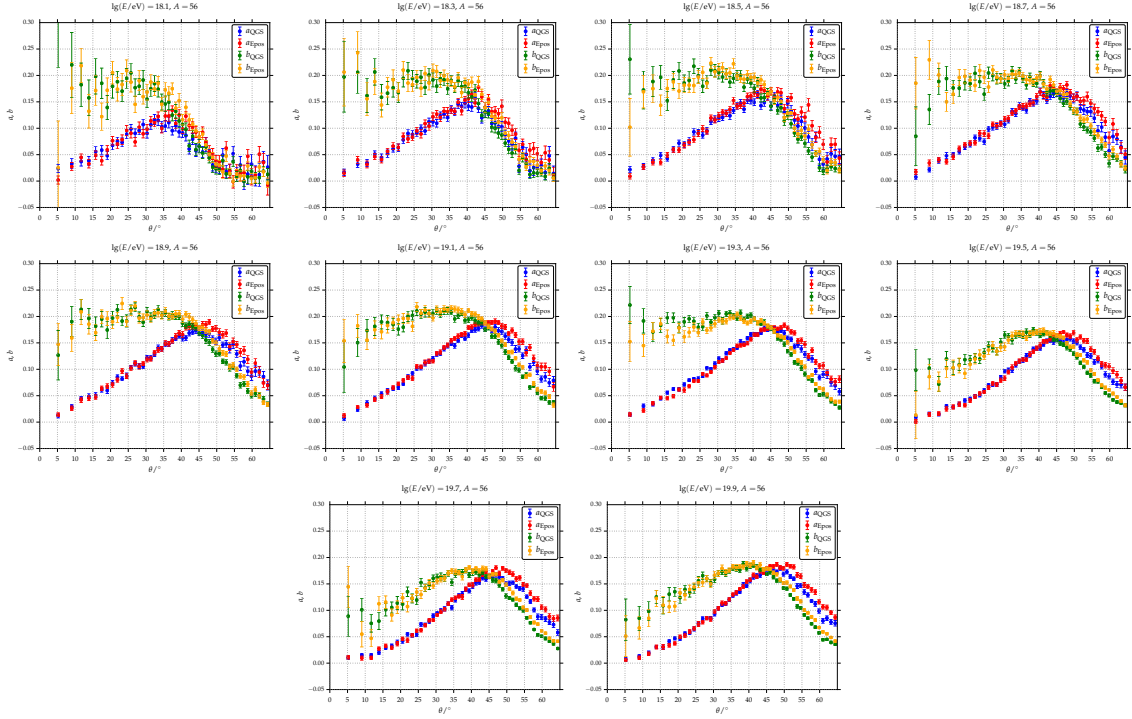


Figure 8: Plots of the asymmetry parameters a and b as a function of the zenith angle θ for iron as the primary particle and for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

Appendix B

The following plots show the dependency of the asymmetry parameters a and b on $\sin^2 \theta$ as well as θ for increasing energy, with respect to the four primaries, proton, helium, oxygen, and iron. The energy ranges form $\lg(E/\text{eV}) = 18.1$ to 19.9 in steps of 0.2 . The red dots represent proton, the blue ones iron, and the grey ones helium and oxygen. Such a collection of plots is shown for both of the hadronic interaction models, QGSJET and EPOS.

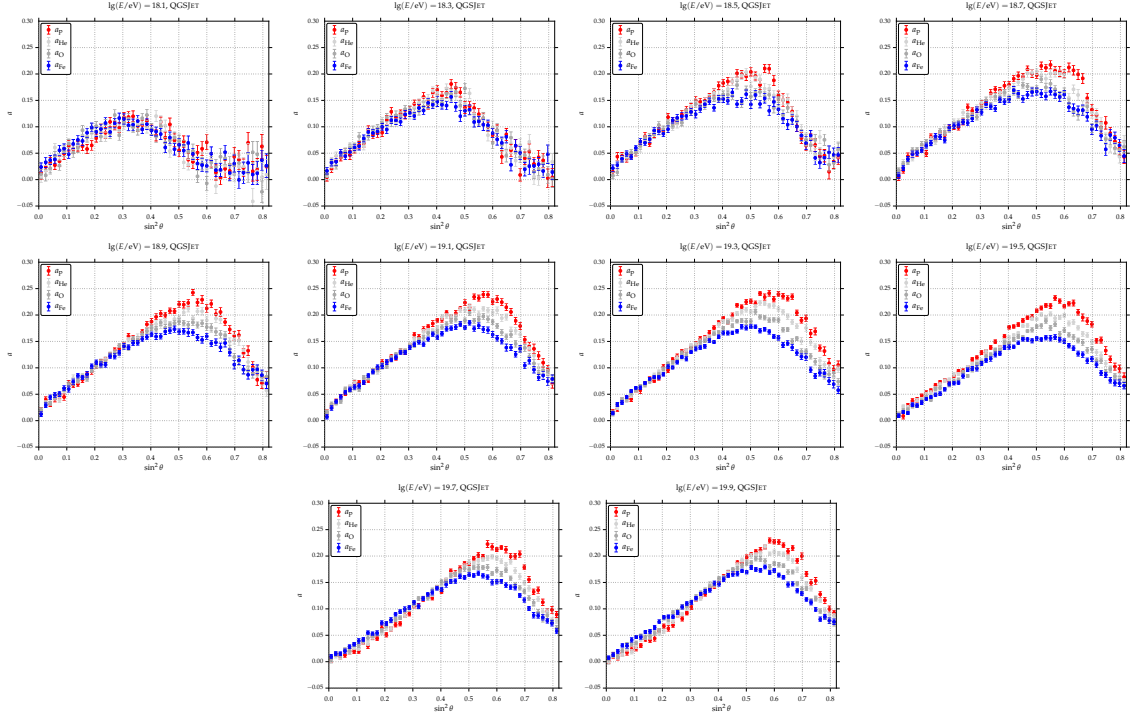


Figure 9: Plots of the dependency of the asymmetry parameter a on $\sin^2 \theta$ in the QGSJET model for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

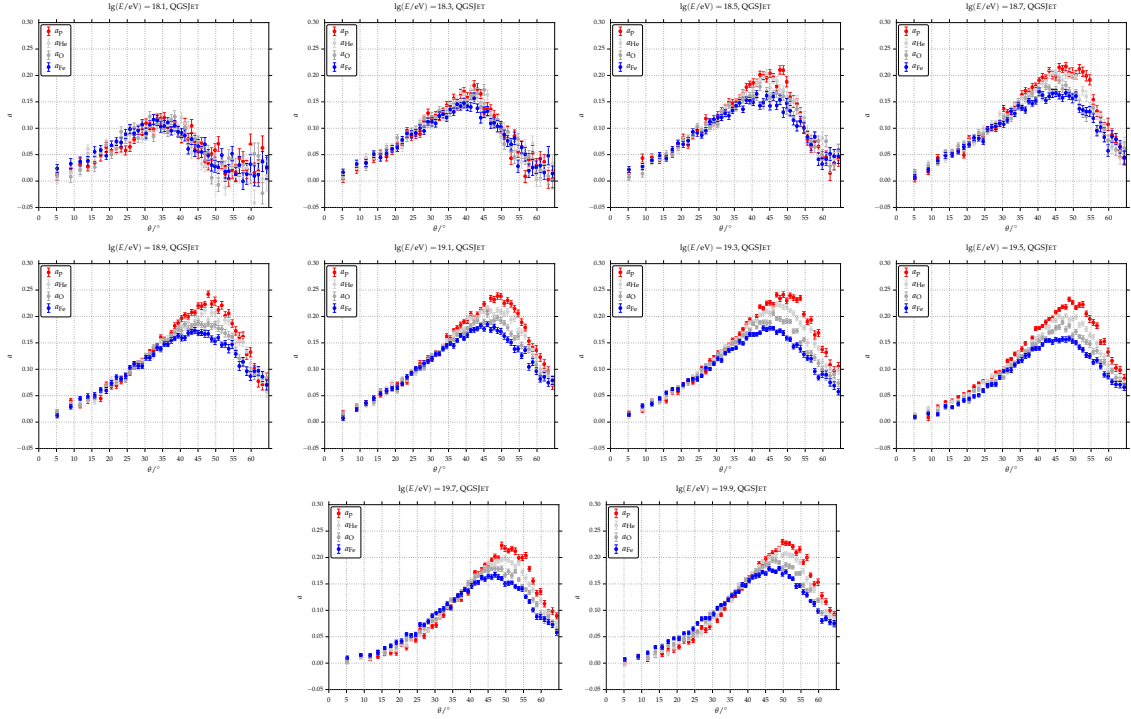


Figure 10: Plots of the dependency of the asymmetry parameter a on θ in the QGSJET model for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

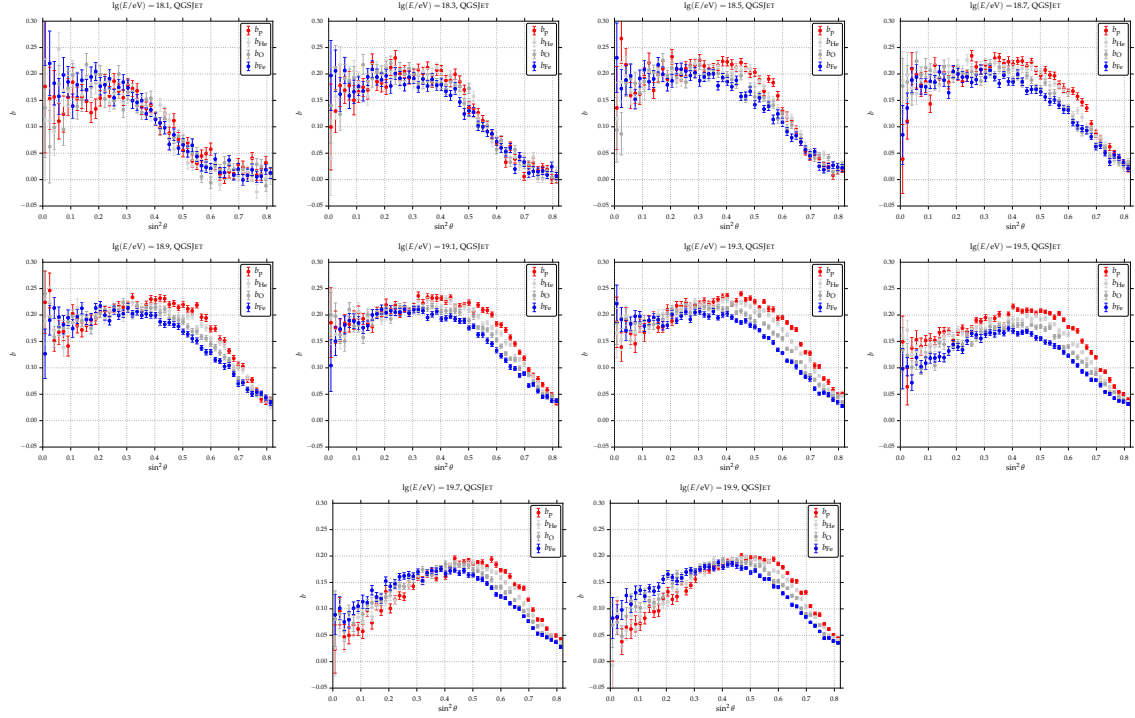


Figure 11: Plots of the dependency of the asymmetry parameter b on $\sin^2 \theta$ in the QGSJET model for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

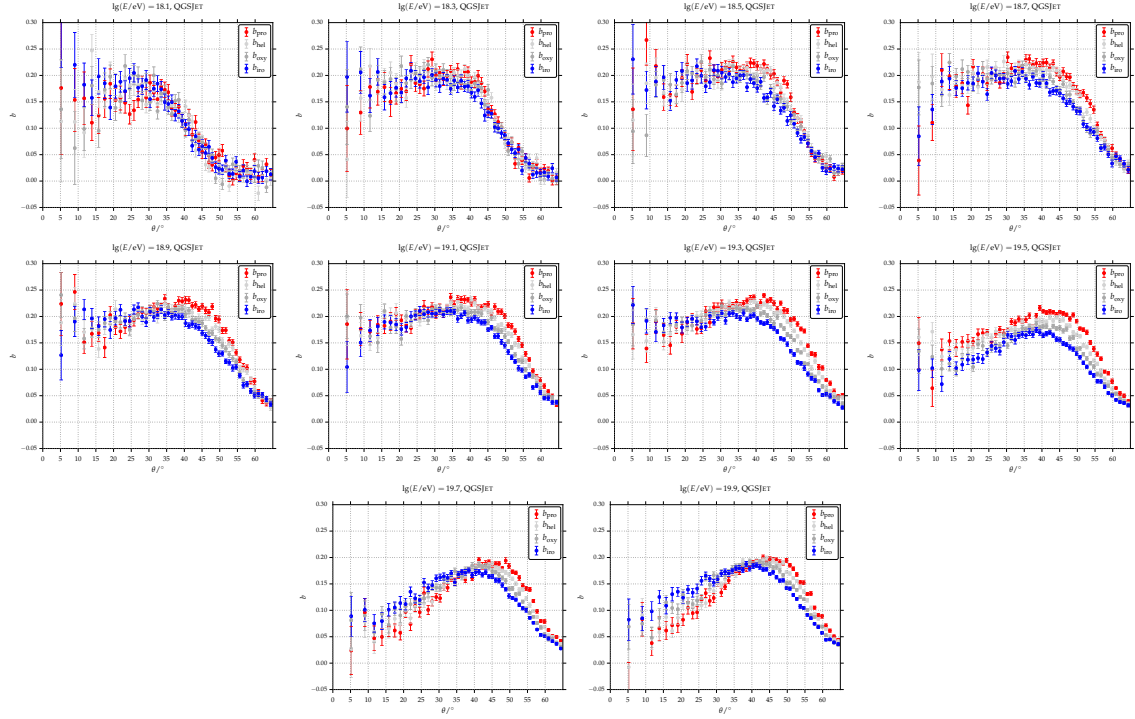


Figure 12: Plots of the dependency of the asymmetry parameter b on θ in the QGSJET model for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

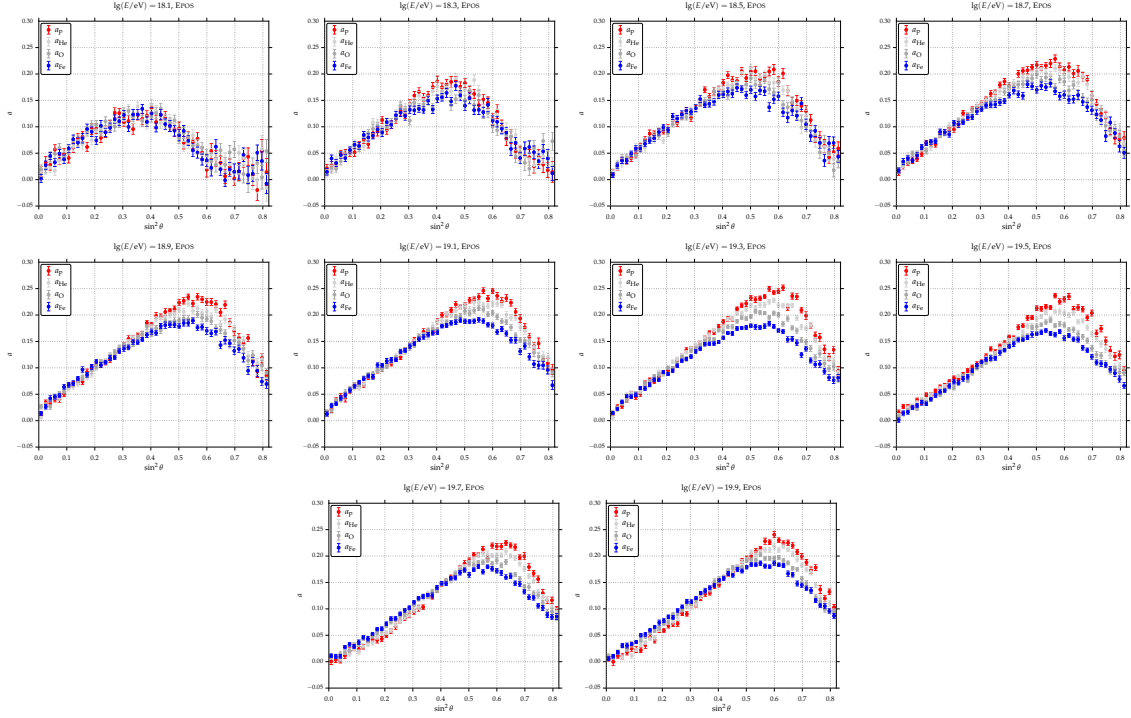


Figure 13: Plots of the dependency of the asymmetry parameter a on $\sin^2 \theta$ in the Epos model for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

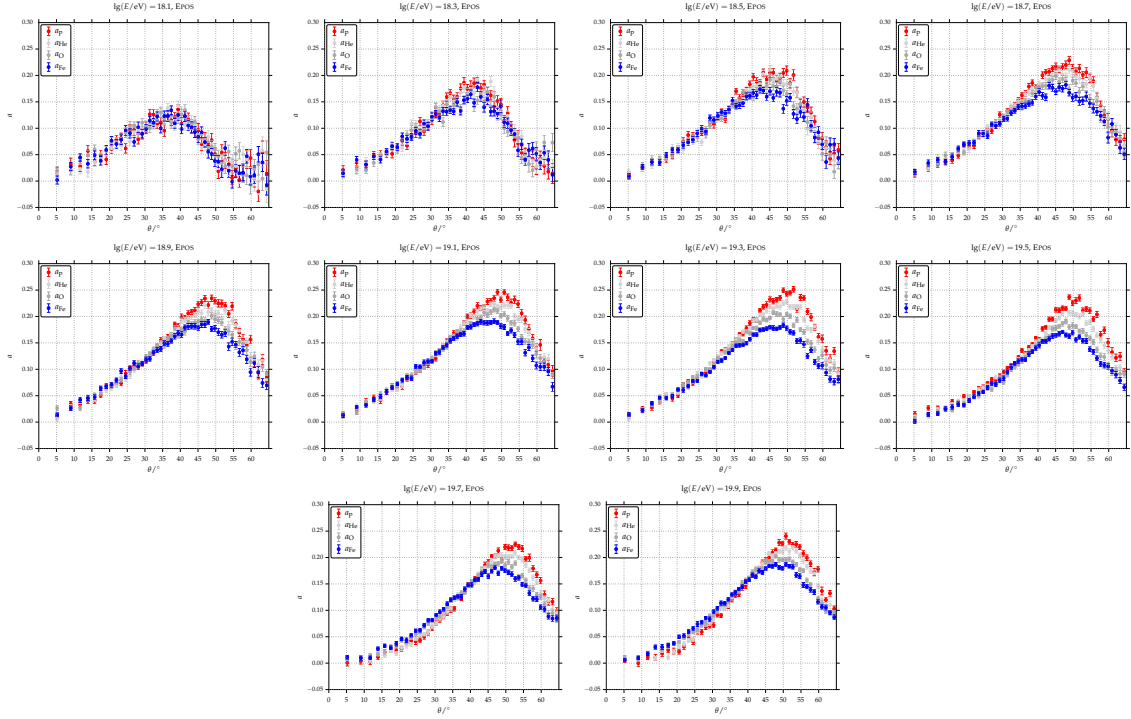


Figure 14: Plots of the dependency of the asymmetry parameter a on θ in the Epos model for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

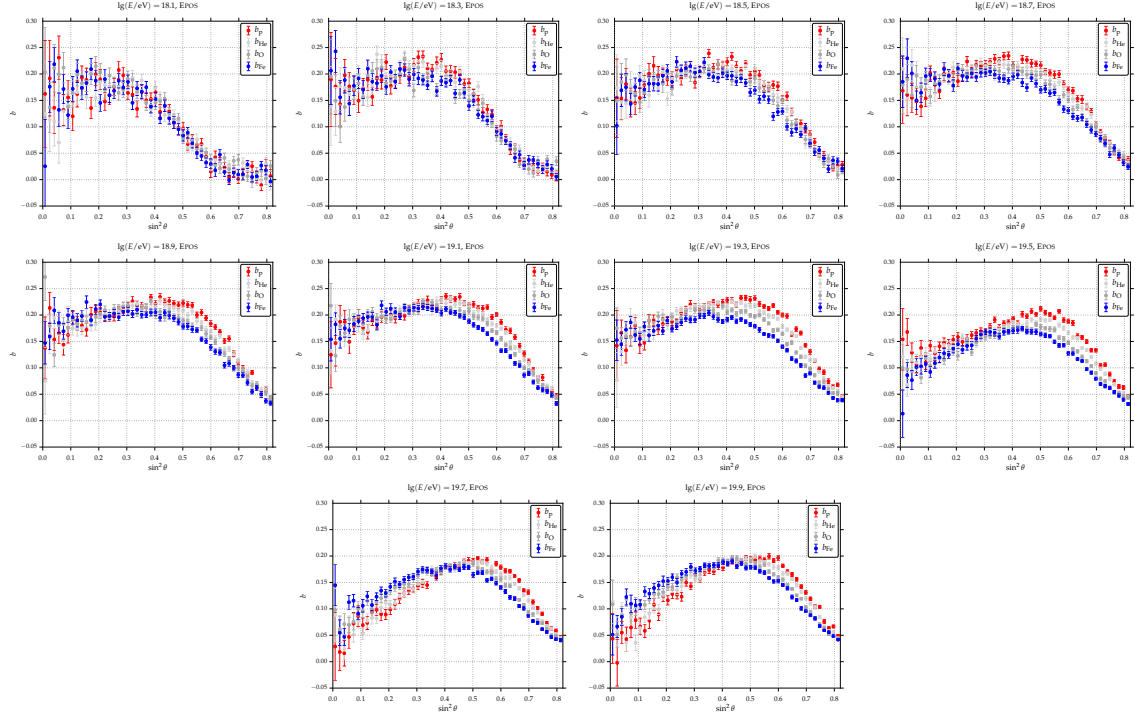


Figure 15: Plots of the dependency of the asymmetry parameter b on $\sin^2 \theta$ in the Epos model for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

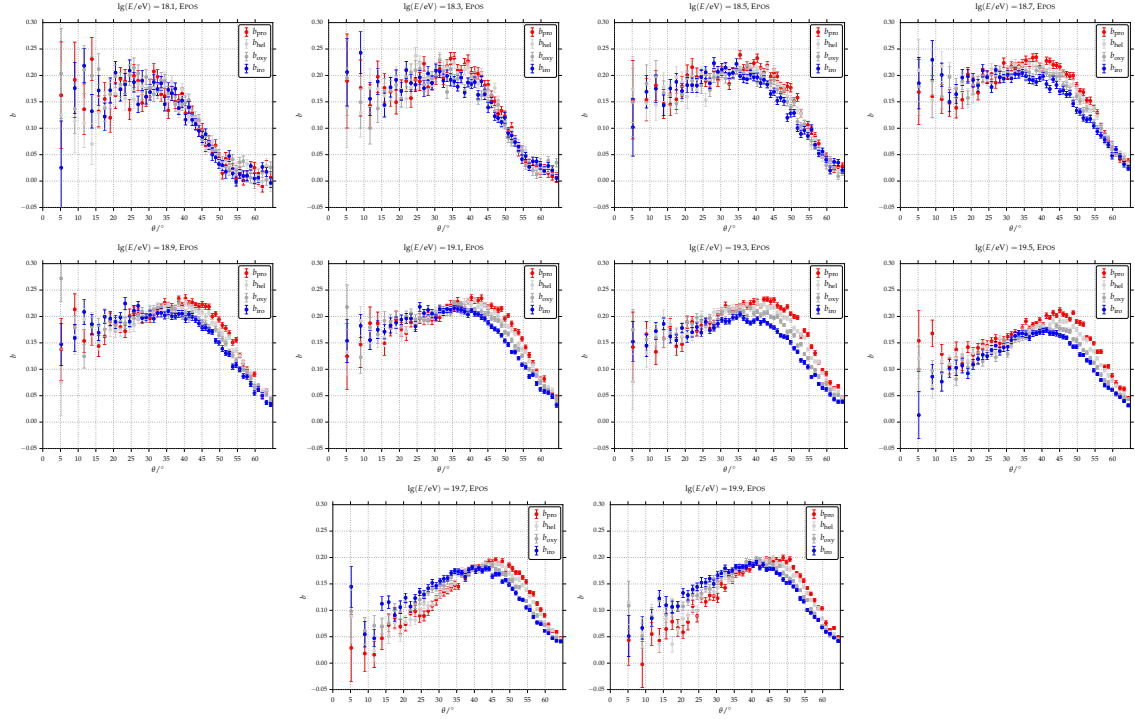


Figure 16: Plots of the dependency of the asymmetry parameter b on θ in the Epos model for increasing energy bins in the range of $\lg(E/\text{eV}) \in [18.1, 19.9]$.

Appendix C

The following plots show the dependency of the asymmetry parameters b in the same way as in Appendix B. However, the empirical function

$$b = (k_0 + k_1 x) \text{logistic}(x; x_0, x_1),$$

with general fit parameters k_0 , k_1 , x_0 , and x_1 is fitted to the proton and to the iron data. The fit function consists of a linear part for small x and a logistic part for larger x .

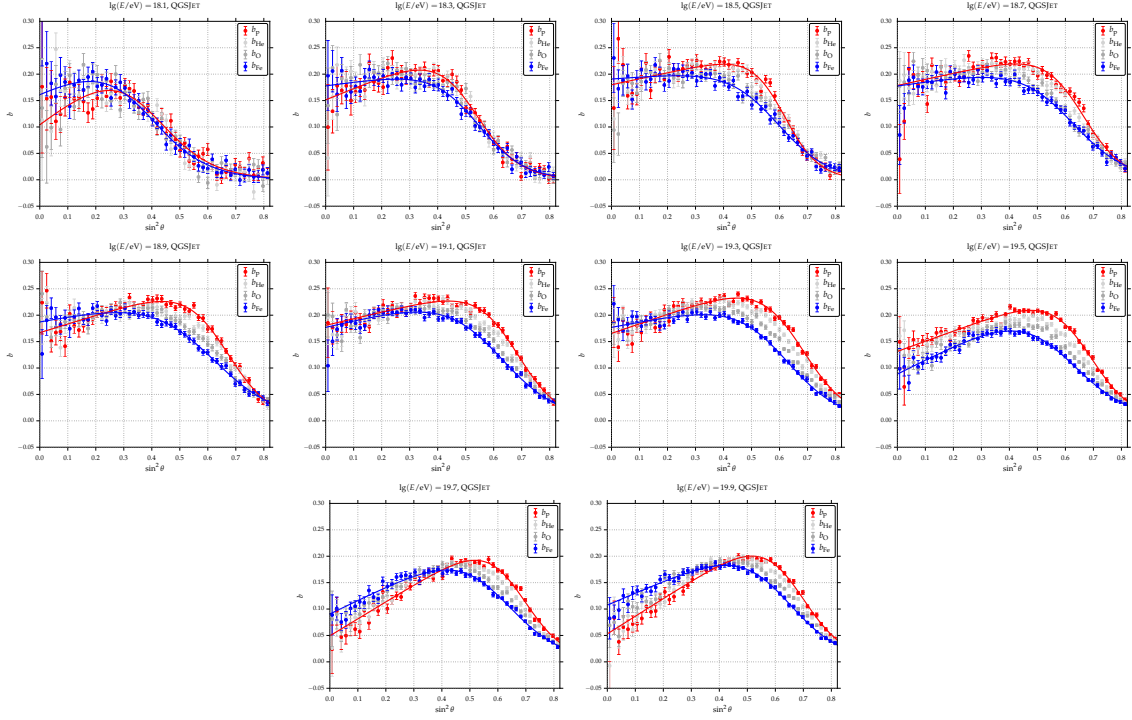


Figure 17: The eight plots show the same dataset as figure 11 fitted with the empirical function $b = (k_0 + k_1 x) \text{logistic}(x; x_0, x_1)$ with general fit parameters k_0 , k_1 , x_0 , and x_1 .

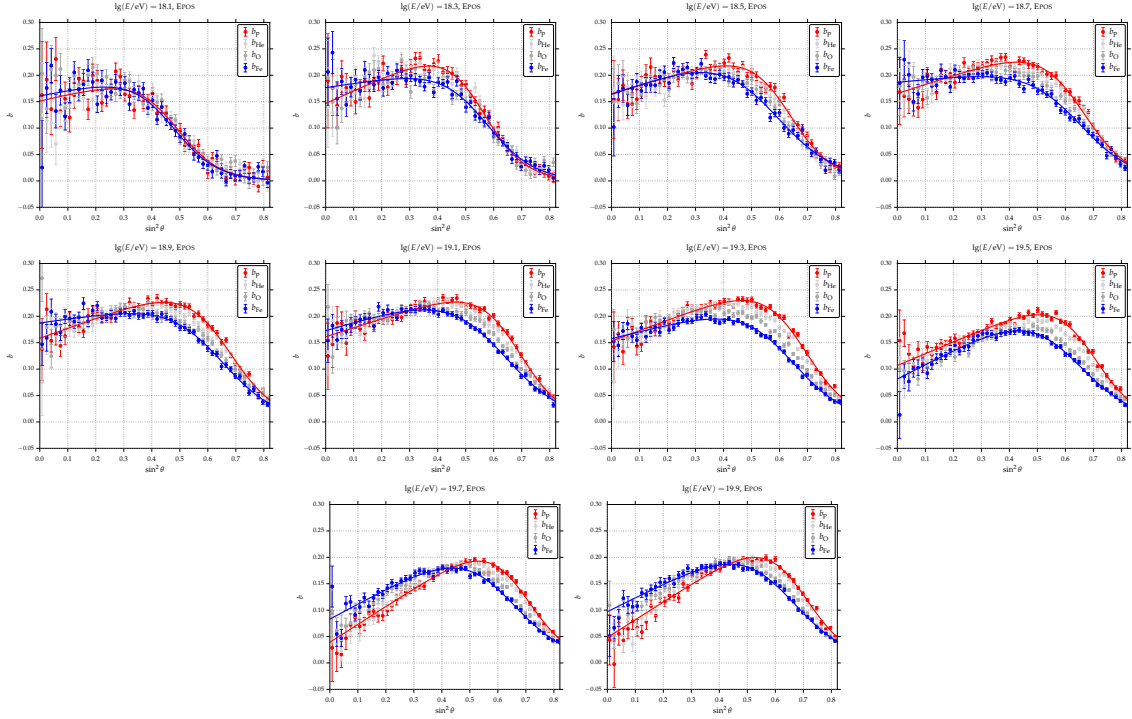


Figure 18: The eight plots show the same dataset as figure 15 fitted with the empirical function $b = (k_0 + k_1 x) \text{logistic}(x; x_0, x_1)$ with general fit parameters k_0 , k_1 , x_0 , and x_1 .

Appendix D

Contours of constant radius r (black), constant apex angle α (red), and constant signal S (blue) are shown in the ground plane (left) and in the shower plane (right) for increasing apex distance d , zenith angle $\theta = 50^\circ$, “neutral” radius $r_y = 1000$ m, attenuation length $\lambda = 36$ km, and power-law exponent $\gamma = 1.4$. A positive x means upstream direction and a negative x means downstream direction.

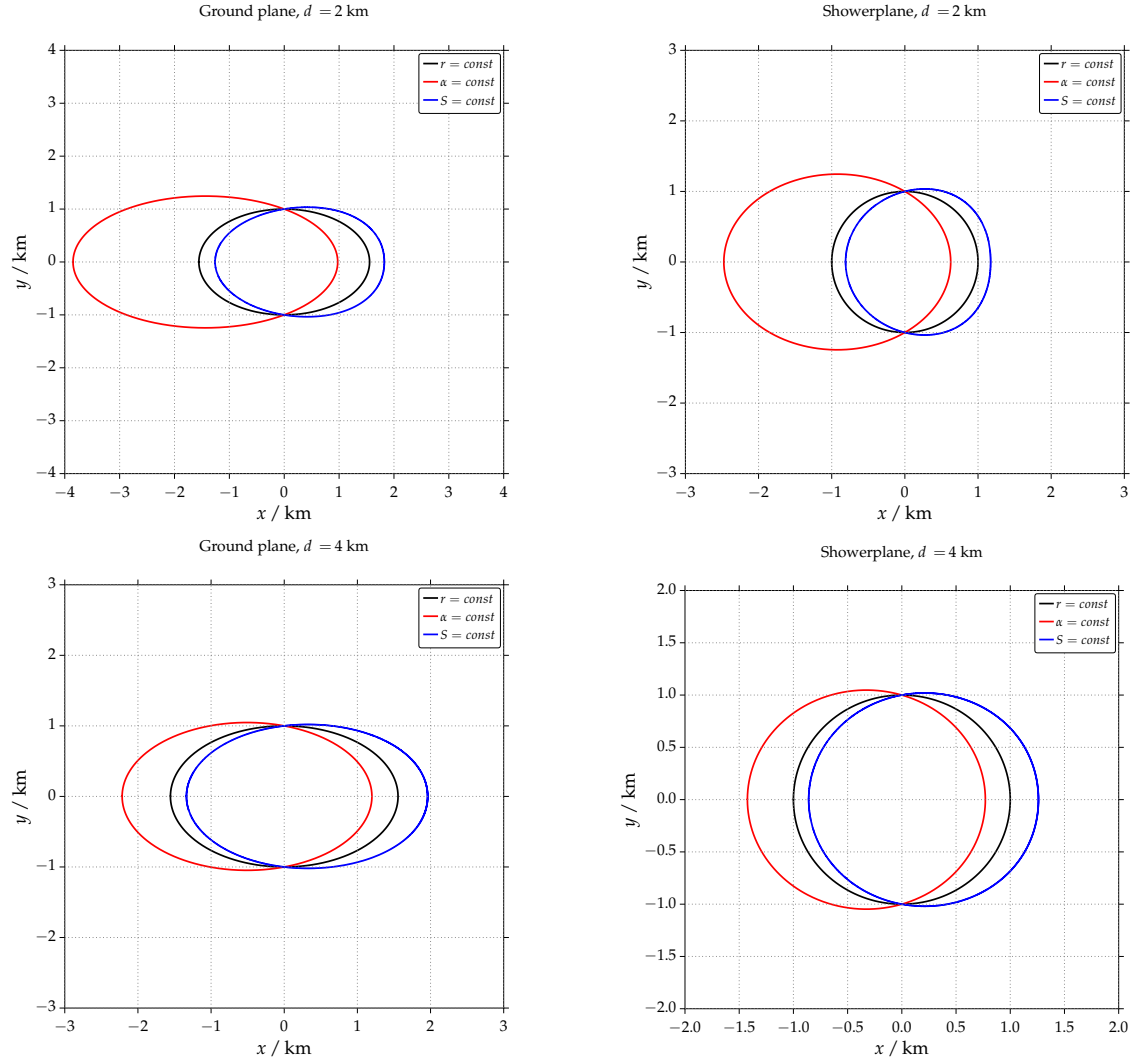


Figure 19: Contours of constant radius r (black), constant apex angle α (red), and constant signal S (blue) in the ground plane (left) and in the shower plane (right).

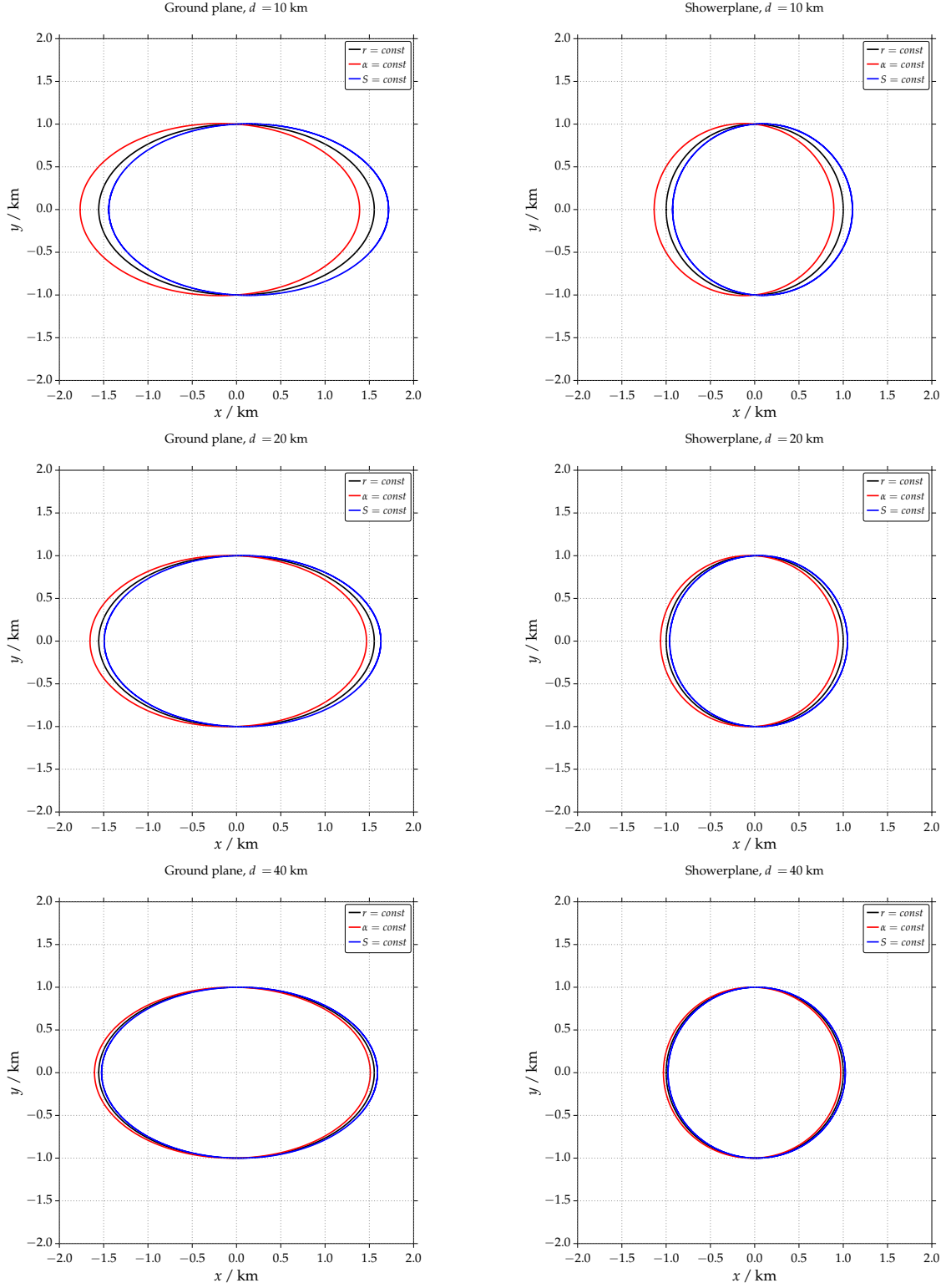


Figure 20: Contours of constant radius r (black), constant apex angle α (red), and constant signal S (blue) in the ground plane (left) and in the shower plane (right).